MOBILE COMPUTING

UNIT I – Introduction: Wireless and Mobile Computing Architecture – Limitations of wireless and mobile communication – Wireless Telecommunication Networks: Digital cellular Systems, TDMA - CDMA – Wireless Networking Techniques –Mobility Bandwidth Tradeoffs – Portable Information Appliances.

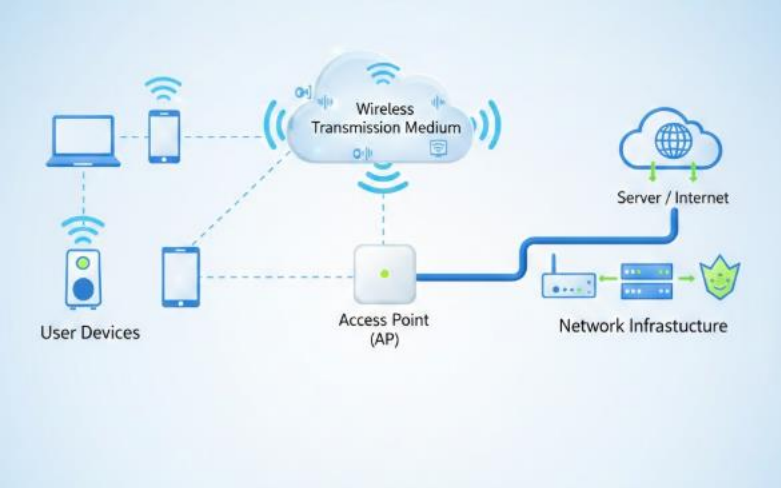
[PAGE NO : 1 TO 18]

Wireless and Mobile Computing Architecture

Wireless Computing

Definition

Wireless Computing refers to a computing environment where devices communicate and access data **without physical wired connections**, using wireless communication technologies.



Architecture of Wireless Computing (in Points)

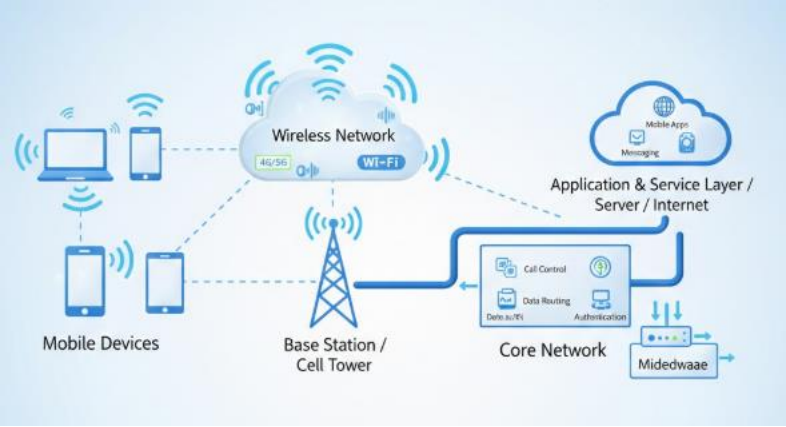
1. **User Devices**
   - Devices such as laptops, desktops, printers, and IoT devices.
   - May be stationary or portable.
   - Access network services wirelessly.
2. **Wireless Transmission Medium**
   - Uses radio waves, infrared, or microwaves.
   - Enables data transfer through air.
   - Supports technologies like Wi-Fi and Bluetooth.

3. **Access Point (AP)**
    - Acts as a central communication hub.
    - Connects wireless devices to the wired network.
    - Manages multiple device connections.
4. **Network Infrastructure**
    - Includes routers, switches, and gateways.
    - Routes data between devices and servers.
    - Provides Internet connectivity.
5. **Server / Internet**
    - Stores data and provides services.
    - Handles requests from wireless devices.
    - Supports applications and cloud services.

Mobile Computing

Definition

Mobile Computing is a type of computing that allows users to access data, applications, and communication services **while moving**, using portable devices connected through wireless networks.



Architecture of Mobile Computing (in Points)

1. **Mobile Devices**
    - Includes smartphones, tablets, PDAs, and laptops.
    - Equipped with sensors and mobility support.
    - Runs mobile operating systems and applications.
2. **Wireless Network**

- Uses cellular networks (3G/4G/5G) and Wi-Fi.
- Provides continuous connectivity during movement.
- Supports roaming and handover.

3. **Base Station / Cell Tower**
   - Provides radio coverage to mobile devices.
   - Manages handoff when users move between cells.
   - Connects devices to the core network.
4. **Core Network**
   - Handles call control, data routing, and mobility management.
   - Performs authentication, authorization, and billing.
   - Connects to public networks and the Internet.
5. **Middleware**
   - Supports data synchronization and session management.
   - Hides network complexity from applications.
   - Ensures seamless service during mobility.
6. **Application & Service Layer**
   - Includes mobile apps and web services.
   - Provides user services like GPS, messaging, and payments.
   - Interacts with backend servers and cloud platforms.

Limitations of wireless and mobile communication

Limitations of Wireless Communication

1. **Slow speed**
   - Wireless connections are usually slower than wired ones, especially when many people use them at the same time.
2. **Signal problems**
   - Walls, buildings, trees, and distance can weaken the signal and cause poor connection.
3. **Interference**
   - Other wireless devices can disturb the signal, which reduces quality and speed.
4. **Security risk**
   - Anyone nearby can try to access the network if it is not well protected.
5. **Limited range**
   - Wireless signals work only within a certain distance from the access point.
6. **Connection not stable**
   - The connection may drop suddenly due to movement or obstacles.

7. **Weather effect**
    - Rain, storms, and bad weather can affect signal quality.
8. **Shared connection**
    - Many users share the same network, which slows it down.
9. **Lower reliability**
    - Wireless networks are more likely to fail than wired networks.
10. **Setup cost**

- Good quality wireless equipment can be expensive.

Limitations of Mobile Communication

1. **Battery drains fast**
    - Using mobile data, calls, and apps uses a lot of battery power.
2. **Poor signal in some areas**
    - Signals may be weak or unavailable in villages, hills, or inside buildings.
3. **Call drops**
    - Calls may disconnect when moving from one place to another.
4. **Slow internet speed**
    - Mobile internet is often slower than Wi-Fi or wired internet.
5. **High data cost**
    - Mobile data plans can be expensive, especially for heavy use.
6. **Limited device power**
    - Mobile phones cannot handle heavy tasks like computers.
7. **Small screen size**
    - Small screens make reading and working difficult for long time.
8. **Data loss**
    - Internet may stop suddenly, causing loss of work or downloads.
9. **Network congestion**
    - Too many users in one area can slow down the network.
10. **Health concerns**

- Long-time use of mobile devices may cause eye strain and stress.

Wireless Telecommunication Networks in Mobile Computing

Meaning

Wireless telecommunication networks are networks that allow **mobile devices** like smartphones and tablets to **communicate and exchange data without using wires**, while users are moving from one place to another.

Wireless Telecommunication Networks Used in Mobile Computing

1. Cellular Networks

- These networks divide large areas into small regions called cells.
- Each cell is controlled by a nearby tower.
- Supports voice calls, messages, and mobile internet.
- Used in everyday mobile communication.

2. Wireless Local Area Network (Wi-Fi)

- Provides wireless internet in a small area like homes, schools, and offices.
- Faster than mobile data but works only within short distance.
- Commonly used for browsing and downloading.

3. Bluetooth Network

- Used for short-distance communication.
- Connects devices like phones, headphones, speakers, and smartwatches.
- Consumes less battery power.

4. Satellite Network

- Uses satellites to provide communication over very large areas.
- Useful in remote locations where mobile towers are not available.
- Expensive and slower compared to other networks.

5. Personal Area Network (PAN)

- Very small network around a person.
- Connects personal devices like phone, laptop, and earbuds.
- Works within a few meters.

6. Metropolitan Area Network (MAN)

- Covers a city or large town.

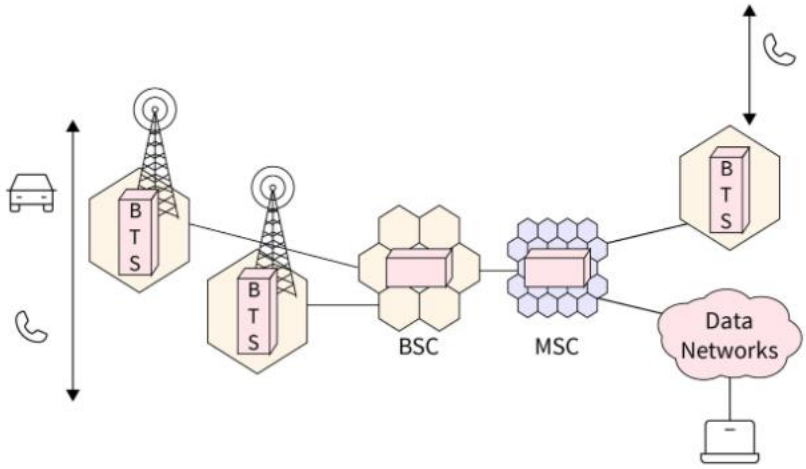
- Used for public wireless internet services.
- Faster than wide area networks.

7. Wide Area Network (WAN)

- Covers very large geographical areas.
- Connects mobile users across countries.
- Used by mobile service providers.

Digital Cellular Systems

- Digital Cellular System is a mobile communication system where information is transmitted in digital form.
- It divides a large geographical area into small regions called **cells**.
- Each cell is served by a fixed **Base Station** for communication with mobile users.
- It supports high capacity by using **frequency reuse technique**.
- It provides better voice quality compared to analog systems.
- It supports services like voice calls, SMS, and mobile internet.
- It improves security using digital encryption techniques.



How it Works

• Area is divided into small cells
• Each cell has a Base Transceiver Station (BTS)
• Mobile connects to nearest BTS when call starts
• BTS sends signal to Base Station Controller (BSC)
• BSC forwards signal to Mobile Switching Center (MSC)

• MSC connects the call to destination user
• When user moves, handoff transfers call without disconnect

Types of Digital Cellular Systems

GSM (Global System for Mobile Communication)

• Definition: GSM is a digital cellular system that uses SIM card to identify users and provides voice, SMS, and internet services
• How it works: Mobile uses SIM to connect to nearest tower, BTS handles signal, and MSC connects the call to destination
• Real-time example: When you use Airtel or Jio SIM to make calls, send SMS, or use mobile data

CDMA (Code Division Multiple Access)

• Definition: CDMA is a system where multiple users share same frequency using unique digital codes
• How it works: All users transmit at same time but system separates signals using codes
• Real-time example: Old Reliance CDMA network used this system for calling

TDMA (Time Division Multiple Access)

• Definition: TDMA is a system where same frequency is divided into time slots for users
• How it works: Each user gets a fixed time slot to send and receive data
• Real-time example: Early GSM mobile networks used TDMA technique

LTE (Long Term Evolution)

• Definition: LTE is a high-speed digital cellular system used for fast internet and data services
• How it works: Data is sent in packet form through high-speed towers with advanced network routing
• Real-time example: 4G internet used for YouTube, Instagram reels, and video calling

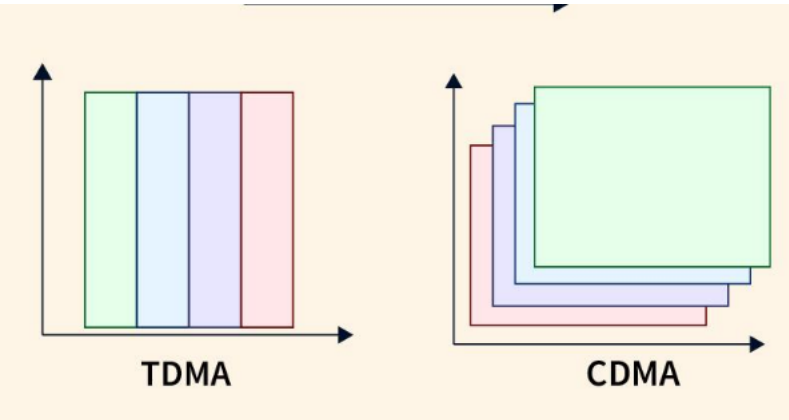
Real-Time Example (Overall)

• When you are talking on mobile while travelling, call does not drop due to handoff
• Network automatically switches towers when you move between cells
• Apps like WhatsApp, Zoom, Google Meet use this system for communication

Applications

• Used in mobile calls and SMS
• Used in internet services like 3G, 4G, 5G
• Used in GPS tracking systems
• Used in emergency services like ambulance and police communication
• Used in online banking and payments
• Used in IoT and smart devices

TDMA – CDMA



TDMA (Time Division Multiple Access)

Definition

• TDMA is a digital multiple access technique
• It allows many users to share the same frequency channel
• Each user is assigned a different time slot for communication
• Users transmit signals one after another in their time slots
• It improves spectrum usage efficiency in mobile networks
• It reduces interference by separating users in time domain

How it Works

• Same frequency channel is divided into small time slots
• Each user gets a fixed time slot to send data

• Only one user transmits at a time in a slot
• Base station combines all time slot signals
• Receiver separates data based on time slots

Real-Time Example

• Early GSM mobile networks used TDMA technique
• When multiple people make calls, each person gets a different time slot on same tower

Applications

• Used in GSM mobile networks (2G systems)
• Used in satellite communication systems
• Used in digital cordless phones
• Used in some broadcasting systems

CDMA (Code Division Multiple Access)

Definition

• CDMA is a digital multiple access technique
• All users share the same frequency at the same time
• Each user is assigned a unique code for communication
• Signals are separated using these digital codes
• It provides high capacity and better security
• It reduces interference compared to other methods

How it Works

• All users transmit data simultaneously
• Each user data is encoded with a unique code
• Receiver uses same code to extract correct signal
• Other signals are treated as noise and ignored
• This allows many users to use same frequency

Real-Time Example

• Reliance CDMA mobile network used this technology
• Many users could call at the same time on same frequency without disturbance

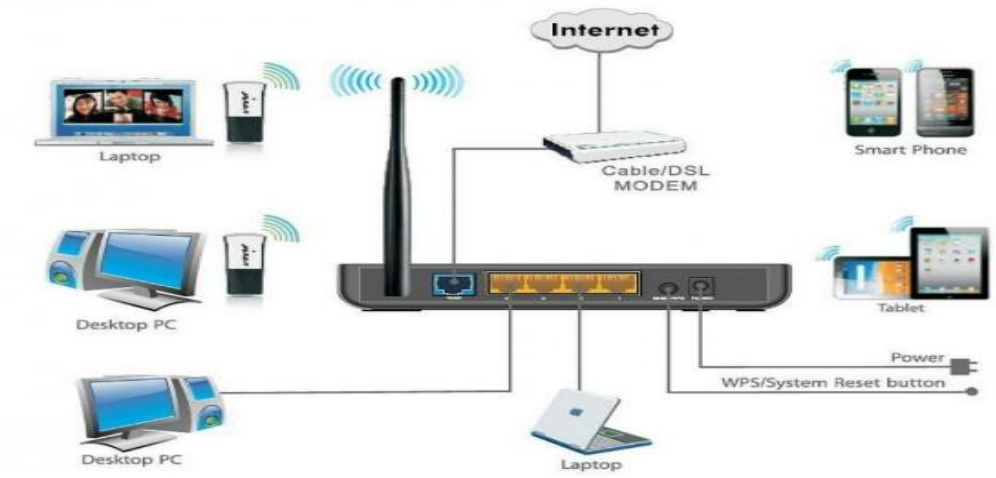
Applications

• Used in CDMA mobile communication systems
• Used in military communication systems
• Used in GPS systems
• Used in wireless data communication

Wireless Networking Techniques

Definition

• Wireless networking techniques are methods used to connect devices without using physical cables
• Communication happens through radio waves, infrared, or microwave signals
• These techniques allow mobility and flexible communication
• They are used in mobile phones, laptops, and IoT devices
• They help in sharing data over short and long distances
• They reduce the need for wired infrastructure



How it Works

• Data is converted into wireless signals
• Signals are transmitted using antennas or access points
• Receiver device captures the signals using its antenna
• Data is decoded back into original form

• Communication happens through wireless medium like air
• Network devices manage signal routing and connection

Types of Wireless Networking Techniques

• Wi-Fi (Wireless Fidelity)

• Definition: Wi-Fi is a wireless technology used for local area networking (LAN)
• How it works: Router sends wireless signals, devices connect using SSID and password
• Real-time example: Internet connection at home, college, or café using Wi-Fi

• Bluetooth

• Definition: Bluetooth is a short-range wireless technology used for device-to-device communication
• How it works: Devices pair and exchange data using radio waves
• Real-time example: Connecting mobile to earbuds or transferring files

• Infrared Communication

• Definition: Infrared is a short-range wireless communication using light signals
• How it works: Devices must be in direct line of sight to communicate
• Real-time example: TV remote control sending signals to TV

• Cellular Networks

• Definition: Cellular networks provide wide-area wireless communication using base stations
• How it works: Mobile connects to nearest tower, data is routed through network
• Real-time example: Mobile calls and 4G/5G internet services

• Satellite Communication

• Definition: Satellite communication uses satellites to transmit signals over long distances
• How it works: Signal is sent to satellite, then retransmitted to earth station
• Real-time example: GPS navigation and DTH TV services

Real-Time Example

• When you connect mobile to Wi-Fi, data is transmitted wirelessly from router to phone
• When you use Bluetooth earphones, sound is transferred without wires
• When you make a mobile call, cellular network connects you through towers

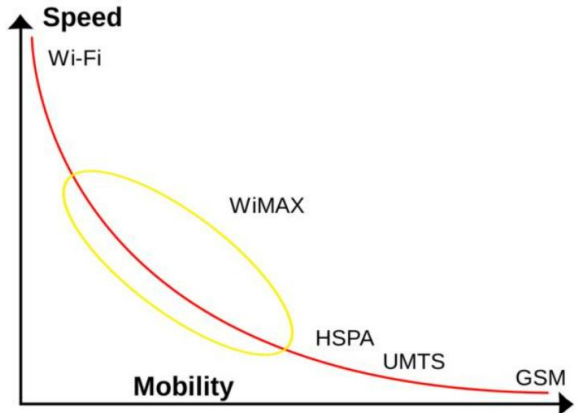
Applications

• Used in mobile communication systems
• Used in internet access (Wi-Fi networks)
• Used in GPS and navigation systems
• Used in smart home devices and IoT
• Used in wireless audio devices like earbuds
• Used in satellite communication and broadcasting

Mobility–Bandwidth Tradeoffs

Definition

• Mobility–Bandwidth tradeoff is a concept in wireless communication
• It explains the balance between user mobility and available bandwidth
• Higher mobility usually reduces available bandwidth efficiency
• Higher bandwidth systems often support lower mobility or shorter range
• It is important in designing wireless networks like 4G, 5G, and Wi-Fi
• It helps in deciding network performance based on user movement and speed



How it Works

• When a user moves faster, network must frequently switch towers (handoff increases)
• This increases signal management overhead in the network

• Due to this, effective bandwidth available for data reduces
• When user is stationary, network gives more stable and higher bandwidth
• High mobility systems need strong infrastructure for continuous connection
• So network design balances speed of movement and data capacity

Types (Based on Tradeoff Levels)

• High Mobility – Low Bandwidth

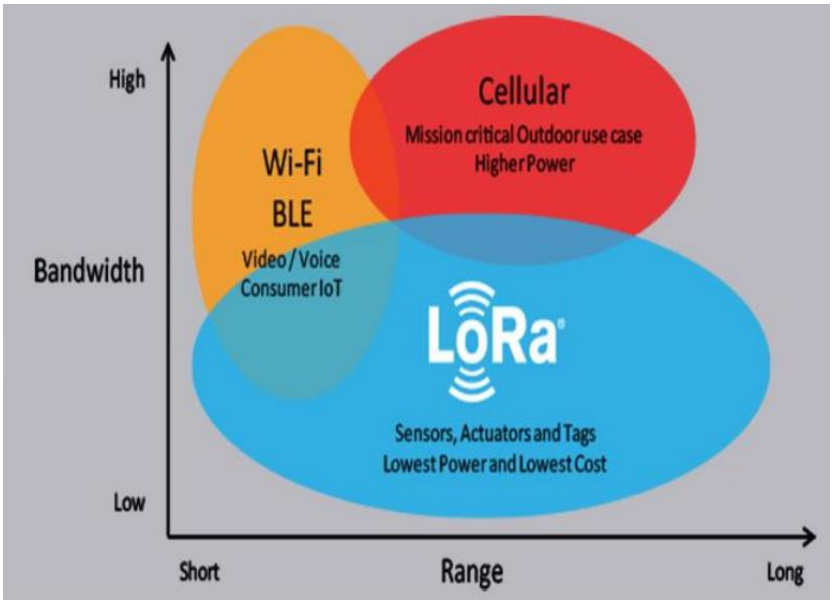
• Definition: Supports users moving at high speed but provides lower data speed
• How it works: Network focuses more on maintaining connection than speed
• Real-time example: Mobile calls during travelling in train or bus

• Low Mobility – High Bandwidth

• Definition: Provides high-speed data but supports limited movement
• How it works: Devices stay mostly fixed for stable high data transmission
• Real-time example: Wi-Fi at home or office used for streaming or downloads

• Balanced Mobility–Bandwidth

• Definition: Provides medium mobility and medium data speed
• How it works: Network adjusts both movement and data transfer efficiently
• Real-time example: Using 4G internet while walking or travelling slowly



Real-Time Example

• When you are in a moving train, internet speed may reduce due to frequent tower switching
• When you sit in one place using Wi-Fi, you get faster and stable internet
• Video calls may lag during high movement but work smoothly in stationary conditions

Applications

• Used in mobile network design (3G, 4G, 5G systems)
• Used in Wi-Fi and hotspot planning
• Used in satellite communication systems
• Used in designing IoT and smart devices
• Used in optimizing video streaming and calling services

Portable Information Appliances

1. Mobile Communication

- Used for voice calls and video calls
- Supports SMS and instant messaging
- Works through cellular networks (4G/5G)
- Enables global communication
- Provides real-time connectivity

How it works

The mobile connects to the nearest base station using cellular signals. Data is transmitted through the telecom network to another user's device.

Real-time example

Calling a friend on WhatsApp or making a normal phone call.

2. Internet Browsing

- Access to websites and online information
- Supports search engines like Google
- Enables social media usage
- Works via Wi-Fi or mobile data
- Provides real-time updates

How it works

The device sends a request through the internet browser, and servers return the requested web pages.

Real-time example

Browsing news on Google Chrome or watching YouTube videos.

3. Online Education

- Supports e-learning platforms
- Live and recorded classes available
- Easy access to study materials
- Interactive learning with apps
- Useful for remote education

How it works

Video conferencing apps connect teachers and students through internet servers in real time.

Real-time example

Attending classes on Zoom or Google Meet.

4. Online Banking & Payments

- Digital money transfers
- Bill payments and recharges
- Secure authentication systems
- 24/7 banking access
- Reduces need for physical banks

How it works

Apps connect to bank servers using encrypted networks to process transactions securely.

Real-time example

Paying using Google Pay or PhonePe in shops.

5. Entertainment Services

- Streaming movies and videos
- Music listening apps
- Mobile gaming support
- On-demand content access
- Personalized recommendations

How it works

Content is streamed from cloud servers to the device using internet bandwidth.

Real-time example

Watching Netflix or listening to Spotify.

6. Healthcare Monitoring

- Tracks heart rate and oxygen levels
- Monitors sleep patterns
- Fitness tracking features
- Remote patient monitoring
- Health alerts and notifications

How it works

Sensors collect body data and send it to mobile apps for analysis.

Real-time example

Using a smartwatch to track daily steps.

7. Navigation & GPS Services

- Real-time location tracking
- Route guidance and directions
- Traffic updates
- Location sharing
- Emergency location tracking

How it works

GPS satellites send signals to the phone, which calculates the exact location.

Real-time example

Using Google Maps while traveling.

8. Business Communication

- Email communication
- Virtual meetings
- File sharing
- Remote working support
- Team collaboration tools

How it works

Apps use internet servers to connect employees in different locations.

Real-time example

Attending office meetings on Microsoft Teams.

9. Social Media Usage

- Instant messaging
- Photo and video sharing
- Live streaming
- Social networking
- Global connectivity

How it works

User data is uploaded to cloud servers and shared with other users instantly.

Real-time example

Chatting on WhatsApp or posting on Instagram.

10. Smart Home Control

- Controls home appliances remotely
- Supports IoT devices

- Improves energy efficiency
- Enhances home security
- Automation features

How it works

Mobile apps send commands through the internet to smart devices at home.

Real-time example

Turning on AC using a mobile app.

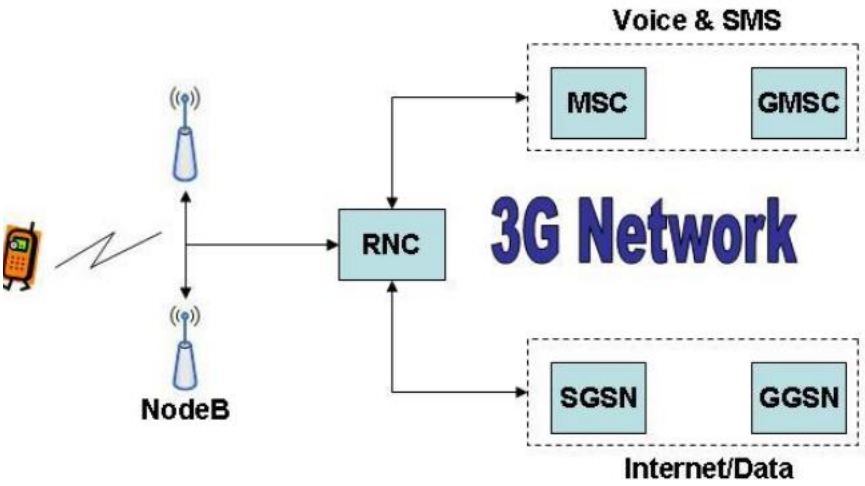
UNIT II – Emerging Wireless Network Standards: 3 G Wireless Networks – State of Industry – Mobility support Software – End User Client Application – Mobility Middleware –Middleware for Application Development - Adaptation and Agents - Service Discovery Middleware - Finding Needed Services - Interoperability and Standardization.

[PAGE NO : 19 TO 50]

3G Wireless Networks

1. Introduction

- 3G stands for Third Generation Wireless Network technology.
- It is the third generation of mobile communication systems developed to provide high-speed data transmission and multimedia services.
- 3G networks support voice communication, video calling, internet browsing, online gaming, multimedia messaging, and mobile TV.
- Compared to 2G networks, 3G provides faster data transfer rates and better network quality.



Features of 3G Wireless Networks

- High-speed internet access.
- Supports multimedia applications.
- Provides video calling facility.
- Better voice quality and network coverage.
- Supports global roaming.
- Higher data transmission speed than 2G.

Architecture of 3G Wireless Networks

1. Mobile Station (MS)

- Mobile station refers to user devices such as smartphones and tablets.
- It allows users to communicate with the network wirelessly.

2. Base Station Subsystem (BSS)

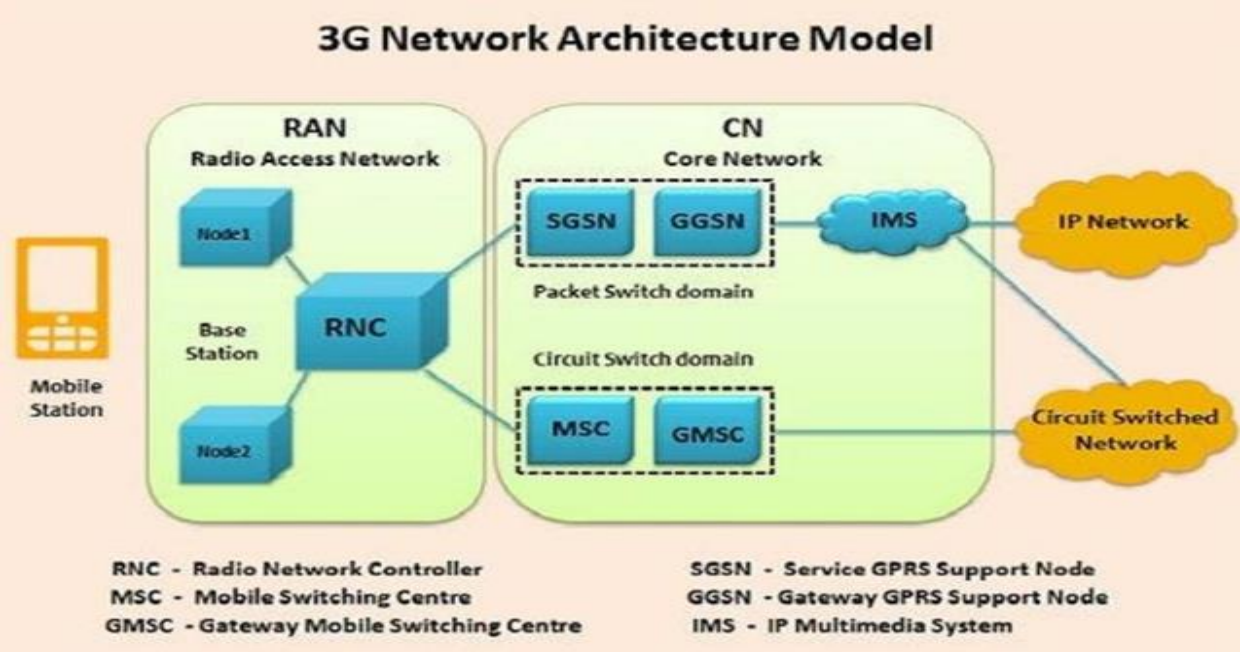
- Base stations provide wireless connectivity between users and the network.
- They manage signal transmission and reception.

3. Radio Network Controller (RNC)

- Controls multiple base stations.
- Manages handoff, radio resources, and network traffic.

4. Core Network (CN)

- Responsible for switching, routing, and internet connectivity.
- Connects users to external networks.



Technologies Used in 3G Networks

1. WCDMA (Wideband Code Division Multiple Access)

- Uses wide bandwidth for high-speed communication.
- Commonly used in UMTS networks.

2. CDMA2000

- Improved version of CDMA technology.
- Supports faster data communication.

3. EDGE

- Enhanced data rates for GSM evolution.
- Improves internet speed in GSM networks.

Services Provided by 3G Networks

1. Video Calling

- Users can communicate using live video and audio simultaneously.

2. Mobile Internet

- High-speed internet browsing and online services are supported.

3. Multimedia Messaging

- Supports transfer of images, audio, and video messages.

4. Mobile TV and Streaming

- Users can watch videos and TV channels online.

5. Online Gaming

- Supports real-time multiplayer gaming through internet connectivity.

Advantages of 3G Wireless Networks

- Faster internet speed compared to 2G.
- Better voice and video quality.
- Supports multimedia applications.
- Improved network capacity and coverage.
- Enables mobile broadband services.

Limitations of 3G Wireless Networks

- Higher power consumption.
- Expensive infrastructure setup.
- Lower speed compared to 4G and 5G.
- Signal quality may reduce in crowded areas.

Applications of 3G Wireless Networks

- Mobile banking.
- Online education.
- Video conferencing.
- GPS and navigation systems.
- Social media and entertainment applications.

Real-Time Example

- Smartphones using 3G networks can perform video calls, browse the internet, stream videos, and access online applications with faster speed compared to traditional 2G networks.

State of Industry

1. Introduction

- State of Industry refers to the current condition, growth, trends, technologies, challenges, and development of industries in a particular sector.
- It shows how industries are adopting modern technologies, improving productivity, and competing in the market.
- In software and communication industries, the state of industry mainly focuses on automation, digital transformation, networking, cloud computing, and wireless technologies.

Features of Modern Industry

- Increased use of automation and artificial intelligence.
- Rapid growth of internet and wireless communication.
- Adoption of cloud computing and big data technologies.
- Global connectivity and digital services.
- Focus on productivity, quality, and customer satisfaction.

Technologies Used in Modern Industry

1. Artificial Intelligence (AI)

- AI is used for automation, prediction, and intelligent decision-making.
- Industries use AI in robotics, healthcare, and business analytics.

2. Internet of Things (IoT)

- IoT connects devices through the internet for smart communication.
- Used in smart homes, industries, and healthcare systems.

3. Cloud Computing

- Provides storage, software, and services through the internet.
- Reduces infrastructure cost and improves scalability.

4. Wireless Communication

- Technologies like 3G, 4G, 5G, and Wi-Fi improve global communication and internet access.

5. Big Data Analytics

- Helps industries analyze huge amounts of data for better business decisions.

Current Trends in Industry

1. Automation

- Industries use automated machines and robots to improve productivity.

2. Digital Transformation

- Traditional systems are replaced with digital technologies and online services.

3. Remote Work and Online Services

- Companies support work-from-home and cloud-based collaboration systems.

4. Cybersecurity

- Strong security systems are implemented to protect digital information and networks.

5. Smart Manufacturing

- Industries use smart machines and sensors for efficient production.

Challenges Faced by Industries

- Cybersecurity threats and data breaches.
- High implementation and maintenance cost.
- Rapid technological changes.
- Requirement for skilled professionals.
- Network and infrastructure limitations.

Advantages of Modern Industry

- Faster communication and processing.
- Increased productivity and efficiency.
- Better customer services.
- Improved data management and automation.
- Global business connectivity.

Limitations of Modern Industry

- High dependency on technology.
- Security and privacy risks.
- Job reduction due to automation.
- Expensive infrastructure and maintenance.

Applications of Modern Industry

- Banking and financial services.
- Healthcare and telemedicine.
- E-commerce and online shopping.
- Smart transportation systems.
- Educational and communication platforms.

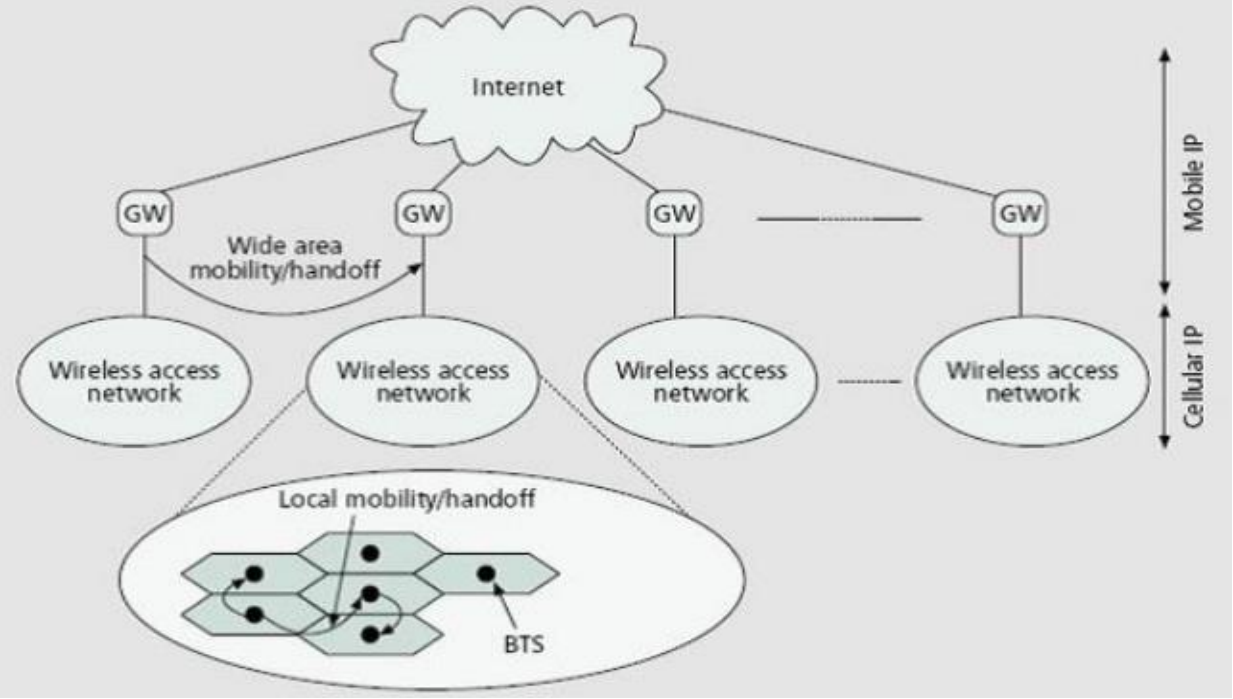
Real-Time Example

- Companies like Amazon and Tesla use AI, cloud computing, IoT, and automation technologies to improve productivity, customer service, and global business operations.

Mobility Support Software

1. Introduction

- Mobility Support Software refers to software systems and technologies that help mobile users communicate and access network services while moving from one location to another.
- It enables seamless connectivity, data transfer, and communication in wireless and mobile networks.
- Mobility support software is widely used in smartphones, wireless networks, and mobile communication systems.



Features of Mobility Support Software

- Supports user mobility and roaming.
- Provides uninterrupted communication during movement.
- Manages handoff between networks.

- Supports wireless internet and multimedia services.
- Ensures secure mobile communication.

Components of Mobility Support Software

1. Mobile Device Software

- Software installed on mobile devices such as smartphones and tablets.
- Manages communication, applications, and wireless connectivity.

Example

- Android and iOS operating systems.

2. Mobility Management Software

- Handles user movement and location tracking in mobile networks.
- Ensures continuous connectivity when users move between network areas.

Functions

- Location management
- Handoff management
- Roaming support

3. Network Support Software

- Software used in wireless networks to manage communication and data transfer.

Functions

- Authentication
- Routing
- Resource allocation

4. Security Software

- Protects mobile communication and user data from unauthorized access.

Functions

- Encryption
- User authentication
- Secure data transfer

Mobility Management Functions

1. Location Management

- Tracks the current location of mobile users.
- Helps deliver calls and messages correctly.

Example

- Mobile network identifying current tower location of a smartphone user.

2. Handoff Management

- Transfers ongoing communication from one base station to another without interruption.

Example

- A phone call continuing smoothly while traveling in a moving vehicle.

3. Roaming Support

- Allows users to access network services outside their home network area.

Example

- Using mobile internet in another city or country.

Advantages of Mobility Support Software

- Enables communication while moving.
- Provides seamless wireless connectivity.
- Supports mobile internet and multimedia applications.
- Improves user convenience and flexibility.
- Supports global communication and roaming.

Limitations of Mobility Support Software

- Security risks in wireless communication.
- High power consumption in mobile devices.
- Network congestion in crowded areas.
- Expensive infrastructure and maintenance.

Applications of Mobility Support Software

- Smartphones and tablets.
- Mobile banking and online payments.
- GPS and navigation systems.
- Video conferencing and online meetings.
- Wireless healthcare monitoring systems.

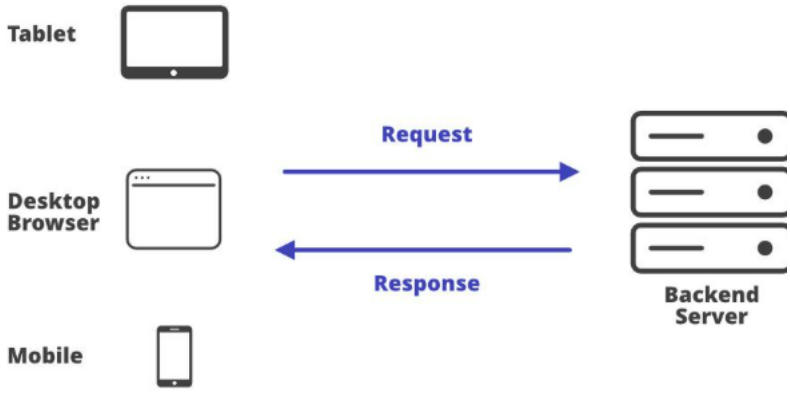
Real-Time Example

- In a smartphone, mobility support software manages location tracking, Wi-Fi and mobile network switching, secure communication, and uninterrupted internet access while users travel between different locations.

End User Client Application

1. Introduction

- End User Client Application refers to software applications that directly interact with users to provide services, communication, and access to system resources.
- These applications run on client devices such as computers, smartphones, tablets, and laptops.
- They provide graphical interfaces and user-friendly features for accessing network and internet services.
- End user applications are commonly used in communication systems, mobile computing, banking, education, and entertainment services.



Features of End User Client Applications

- Provides direct interaction between user and system.
- User-friendly graphical interface.
- Supports network and internet communication.
- Allows data access and processing.
- Supports multimedia and online services.

Components of End User Client Applications

1. User Interface (UI)

- The user interface allows users to interact with the application.
- Includes buttons, menus, forms, and graphical elements.

Example

- Login page of a banking application.

2. Application Logic

- Handles processing and execution of user requests.
- Controls operations and functionality of the application.

Example

- Processing online payment requests in e-commerce applications.

3. Database Connectivity

- Connects the application to databases for storing and retrieving information.

Example

- Retrieving customer details from banking database.

4. Network Communication Module

- Handles communication between client application and server.
- Transfers requests and responses over the network.

Example

- Sending login request from mobile app to server.

5. Security Module

- Protects user data and application access.
- Provides authentication and secure communication.

Example

- Password encryption and OTP verification.

Types of End User Client Applications

1. Desktop Applications

- Applications installed on personal computers or laptops.

Example

- Microsoft Word, VLC Media Player.

2. Web Applications

- Applications accessed using web browsers through the internet.

Example

- Gmail, Online banking websites.

3. Mobile Applications

- Applications designed for smartphones and tablets.

Example

- WhatsApp, Google Pay, Instagram.

4. Cloud-Based Applications

- Applications that use cloud servers for storage and processing.

Example

- Google Drive, Dropbox.

Functions of End User Client Applications

1. Data Processing

- Processes user input and generates required output.

2. Communication

- Supports messaging, calling, and data exchange.

3. Resource Access

- Allows users to access files, databases, and online services.

4. Multimedia Support

- Supports audio, video, and image handling.

5. Authentication

- Verifies user identity for secure access.

Applications of End User Client Applications

- Online banking systems.
- Social media platforms.
- Video conferencing systems.
- E-learning platforms.

- E-commerce applications.

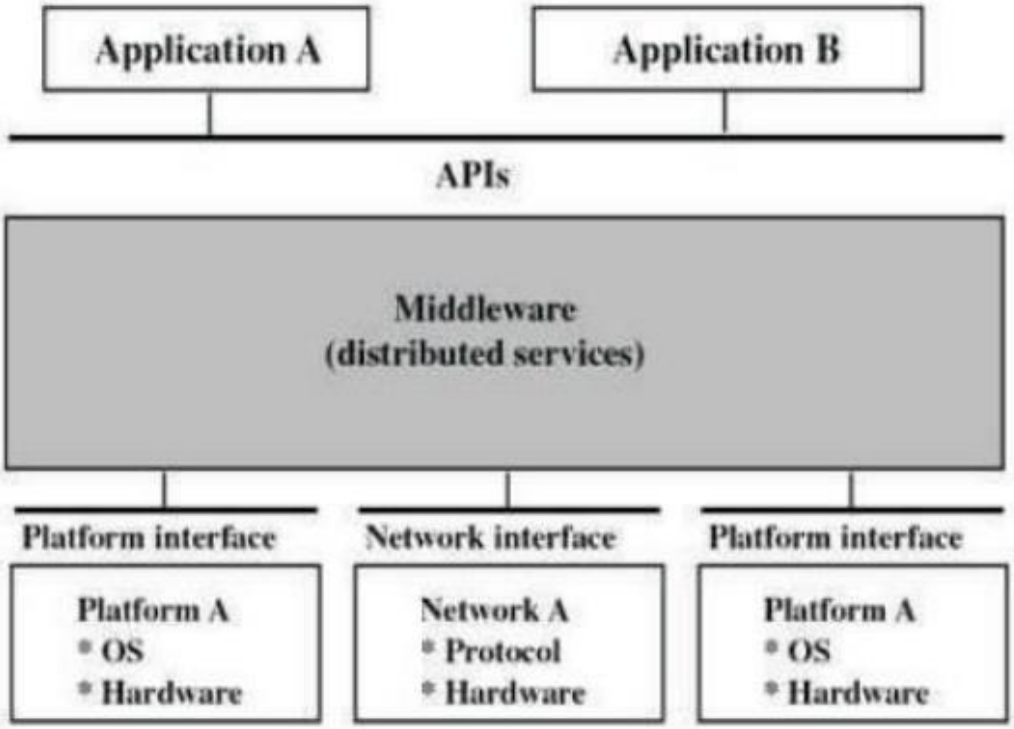
Real-Time Example

- In a mobile banking application, the end user client application allows users to log in, check account balance, transfer money, and pay bills securely through a smartphone connected to the internet.

Mobility Middleware

1. Introduction

- Mobility Middleware is a software layer that provides communication and coordination between mobile applications, mobile devices, and network services.
- It helps mobile users access services and data seamlessly while moving between different networks and locations.
- Mobility middleware hides the complexity of wireless communication and mobility management from applications.
- It supports mobile computing, distributed systems, and wireless network environments.



Components of Mobility Middleware

1. Communication Manager

- Handles communication between mobile devices and network servers.
- Manages message transfer and data exchange.

Example

- Sending requests from mobile banking app to server.

2. Mobility Manager

- Manages user movement and network handoff operations.
- Ensures continuous connectivity during movement.

Functions

- Handoff management
- Roaming support
- Location tracking

3. Synchronization Manager

- Synchronizes data between mobile devices and central servers.
- Ensures updated information is available on all devices.

Example

- Updating cloud files automatically across devices.

4. Security Manager

- Provides authentication, encryption, and secure communication.
- Protects mobile data from unauthorized access.

Example

- OTP verification and encrypted transactions.

5. Resource Manager

- Manages device resources such as memory, battery, and bandwidth efficiently.

Example

- Reducing network usage during low battery conditions.

Functions of Mobility Middleware

1. Mobility Support

- Allows users to move across networks without interrupting communication.

2. Data Management

- Handles storage, retrieval, and synchronization of mobile data.

3. Network Management

- Manages wireless connections and network switching.

4. Security Services

- Provides secure access and communication.

5. Application Support

- Helps mobile applications communicate with distributed systems.

Types of Mobility Middleware

1. Message-Oriented Middleware

- Supports message-based communication between systems.

Example

- Mobile messaging services.

2. Database Middleware

- Provides database connectivity for mobile applications.

Example

- Mobile banking database access.

3. Transaction Middleware

- Handles secure mobile transactions and payment processing.

Example

- Online payment applications.

Applications of Mobility Middleware

- Mobile banking systems.
- E-commerce applications.
- Cloud-based mobile applications.
- GPS and navigation systems.
- Wireless healthcare systems.

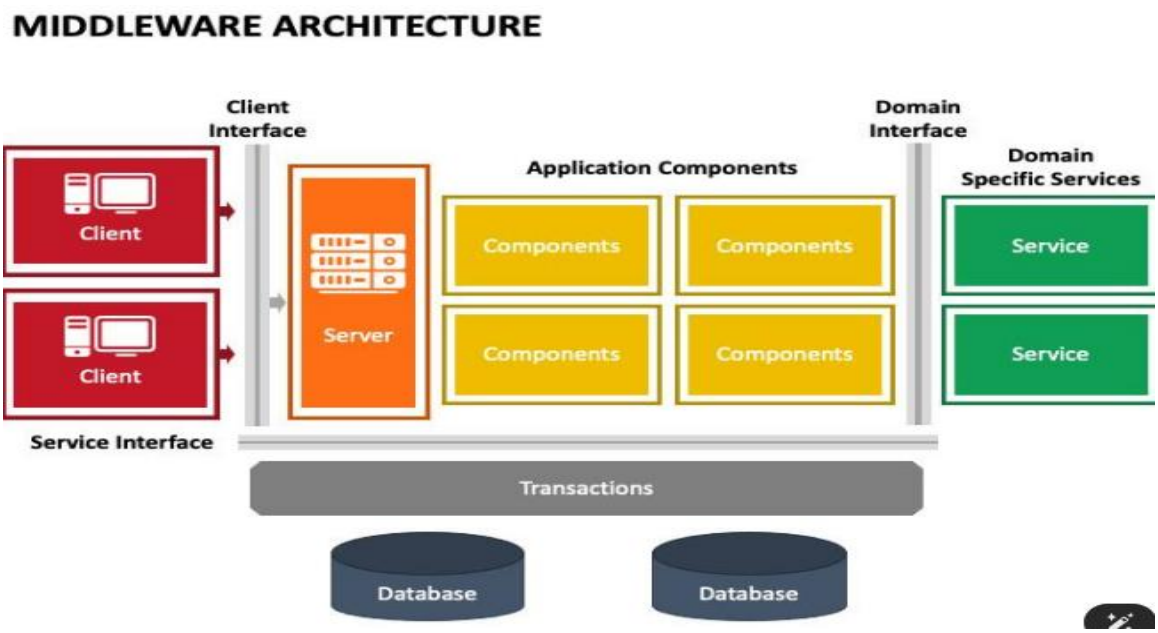
Real-Time Example

- In a mobile banking system, mobility middleware manages communication between smartphone applications and banking servers, supports secure transactions, handles roaming between networks, and synchronizes account data while users move between locations.

Middleware for Application Development

1. Introduction

- Middleware for Application Development is a software layer that acts as a bridge between operating systems, databases, networks, and applications.
- It helps developers build distributed and network-based applications easily by providing common services and communication support.
- Middleware simplifies application development by hiding the complexity of networking, communication, security, and data management.
- It is widely used in web applications, mobile applications, cloud systems, and enterprise software.



Features of Middleware for Application Development

- Supports communication between applications and servers.
- Provides interoperability between different platforms.
- Simplifies distributed application development.
- Supports security and authentication services.
- Handles database connectivity and transaction management.

Components of Middleware for Application Development

1. Communication Middleware

- Provides communication between distributed applications and servers.
- Handles message passing and remote procedure calls.

Example

- Client-server communication in online banking systems.

2. Database Middleware

- Provides connectivity between applications and databases.
- Allows applications to store and retrieve data efficiently.

Example

- JDBC connecting Java applications with MySQL database.

3. Transaction Middleware

- Manages transactions and ensures data consistency during operations.

Example

- Online payment processing systems.

4. Security Middleware

- Provides authentication, authorization, and data encryption.
- Protects applications from unauthorized access.

Example

- OTP verification in banking applications.

5. Web Middleware

- Supports development and execution of web-based applications.

Example

- Apache Tomcat and web servers.

Functions of Middleware for Application Development

1. Application Integration

- Connects different applications and services together.

2. Data Management

- Handles data transfer, storage, and synchronization.

3. Security Management

- Provides secure communication and user authentication.

4. Transaction Processing

- Ensures reliable transaction execution.

5. Resource Management

- Manages system resources like memory and network connections.

Types of Middleware for Application Development

1. Message-Oriented Middleware (MOM)

- Supports communication using messages between distributed systems.

Example

- Email and messaging applications.

2. Object Request Broker (ORB)

- Allows objects in distributed systems to communicate with each other.

Example

- CORBA architecture.

3. Remote Procedure Call (RPC) Middleware

- Allows applications to execute functions on remote systems.

Example

- Client requesting data from server remotely.

4. Database Middleware

- Provides database access and query processing.

Example

- ODBC and JDBC connectivity.

Applications of Middleware for Application Development

- Online banking systems.
- E-commerce applications.
- Cloud computing platforms.
- Enterprise resource planning systems.
- Mobile and web applications.

Real-Time Example

- In an online shopping application, middleware connects the user interface, payment gateway, inventory database, and server communication to ensure smooth processing of orders, payments, and product updates.

Adaptation and Agents

1. Introduction

- Adaptation and Agents are important concepts in mobile computing, distributed systems, and intelligent software systems.
- Adaptation refers to the ability of a system or application to adjust its behavior according to changing environments and user requirements.
- Agents are software entities that perform tasks automatically on behalf of users or systems.
- Together, adaptation and agents improve flexibility, automation, and efficiency in modern applications.

Adaptation

Definition

- Adaptation is the process in which a system modifies its behavior, configuration, or operations according to environmental changes, user needs, or network conditions.

Features of Adaptation

- Supports dynamic environment changes.
- Improves system performance and efficiency.
- Provides personalized services to users.
- Helps systems work under limited resources.

Types of Adaptation

1. User Adaptation

- System changes behavior according to user preferences and requirements.

Example

- Mobile apps changing language and themes based on user settings.

2. Network Adaptation

- System adjusts according to network bandwidth and connectivity conditions.

Example

- Video quality automatically reducing when internet speed becomes slow.

3. Device Adaptation

- Applications adapt according to device capabilities such as screen size, memory, and battery.

Example

- Websites adjusting layout for mobile phones and tablets.

4. Context Adaptation

- System changes behavior based on user location, time, or environment.

Example

- GPS applications suggesting routes based on current traffic conditions.

Applications of Adaptation

- Mobile computing systems.
- Smart devices and IoT systems.
- E-learning applications.
- Streaming and multimedia platforms.

Agents

Definition

- Agents are autonomous software programs that perform tasks automatically for users or systems without continuous human intervention.

Features of Agents

- Autonomous behavior.
- Decision-making capability.
- Communication with other agents or systems.
- Goal-oriented task execution.
- Ability to learn and adapt.

Components of Agents

1. Sensor

- Collects information from environment or system.

Example

- Detecting user location in navigation applications.

2. Decision Module

- Processes collected information and makes decisions.

Example

- Recommending best route in GPS systems.

3. Action Module

- Performs actions based on decisions taken.

Example

- Sending alerts or notifications automatically.

Types of Agents

1. Mobile Agents

- Move from one system or network to another to perform tasks.

Example

- Mobile shopping assistants collecting product information.

2. Intelligent Agents

- Use artificial intelligence techniques for decision making.

Example

- Virtual assistants like Siri and Google Assistant.

3. Collaborative Agents

- Work together with other agents to complete tasks.

Example

- Multi-agent systems in robotics.

4. Reactive Agents

- Respond immediately to environmental changes.

Example

- Smart traffic control systems.

Applications of Agents

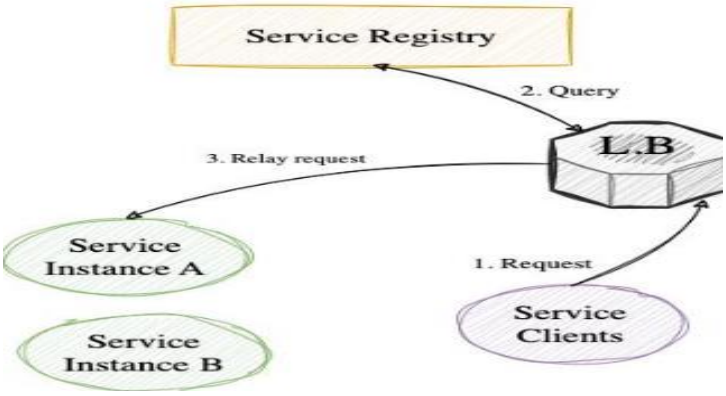
- E-commerce systems.
- Mobile banking applications.
- Smart homes and IoT systems.
- Healthcare monitoring systems.
- Artificial intelligence applications.

Real-Time Example

- In a smartphone navigation system, adaptation adjusts map quality based on internet speed, while intelligent agents automatically suggest shortest routes, traffic updates, and nearby services according to user location.

Service Discovery Middleware

1. Introduction

- Service Discovery Middleware is a software layer that helps devices, applications, and services automatically discover and communicate with each other in distributed and mobile computing environments.
- It removes the need for manual configuration by allowing systems to identify available services dynamically.
- Service discovery middleware is widely used in mobile computing, cloud systems, IoT devices, and distributed networks.
- 

Components of Service Discovery Middleware

1. Service Provider

- A service provider offers services or resources to other devices or applications.

Example

- A printer connected to a wireless network providing printing service.

2. Service Consumer

- A service consumer searches and uses available services.

Example

- A laptop searching for nearby wireless printer service.

3. Service Registry

- Maintains information about available services in the network.
- Stores service details such as name, address, and type.

Example

- Directory storing information about available IoT devices.

4. Discovery Protocol

- Protocol used for searching and locating services automatically.

Example

- Bluetooth service discovery protocols.

Functions of Service Discovery Middleware

1. Service Registration

- Services register themselves in the network when they become available.

Example

- Smart TV registering streaming service in home network.

2. Service Discovery

- Applications or devices search for required services automatically.

Example

- Smartphone searching for nearby Wi-Fi printer.

3. Service Selection

- Best available service is selected based on requirements.

Example

- Selecting nearest available cloud server.

4. Service Communication

- Middleware establishes communication between service provider and consumer.

Example

- Mobile app communicating with payment gateway service.

Types of Service Discovery Middleware

1. Directory-Based Discovery

- Uses a central directory or registry to maintain service information.

Example

- DNS service directories.

2. Directory-Less Discovery

- Devices directly communicate and discover services without central registry.

Example

- Bluetooth device discovery.

Technologies Used in Service Discovery Middleware

1. UPnP (Universal Plug and Play)

- Allows automatic discovery of devices in local networks.

2. Jini

- Java-based service discovery architecture.

3. Bluetooth SDP (Service Discovery Protocol)

- Used for discovering Bluetooth services.

4. Zeroconf

- Enables automatic networking without manual configuration.

Applications of Service Discovery Middleware

- Smart homes and IoT systems.
- Mobile computing applications.
- Cloud computing environments.
- Wireless communication systems.
- Distributed enterprise systems.

Real-Time Example

- In a smart home network, Service Discovery Middleware allows smartphones, smart TVs, printers, and IoT devices to automatically discover and communicate with each other without manual network configuration.

Finding Needed Services

1. Introduction

- Finding Needed Services refers to the process of locating and accessing required services in distributed systems, mobile computing, and network environments.
- It helps users and applications discover available resources and services automatically without manual configuration.
- This process is important in mobile networks, cloud computing, IoT systems, and web services where services may dynamically appear or disappear.

Components Involved in Finding Needed Services

1. Service Provider

- Provides services or resources to users and applications.

Example

- Printer service in a wireless network.

2. Service Consumer

- Searches and accesses required services.

Example

- Laptop searching for nearby printer service.

3. Service Registry

- Stores information about available services in the network.

Example

- Directory containing details of cloud services.

4. Discovery Protocol

- Protocol used to search and locate services automatically.

Example

- Bluetooth Service Discovery Protocol (SDP).

Process of Finding Needed Services

1. Service Registration

- Service providers register their services in the network or service registry.

Example

- Smart TV registering media streaming service.

2. Service Advertisement

- Available services announce their presence to users and systems.

Example

- Wi-Fi printer broadcasting printing service availability.

3. Service Discovery

- Users or applications search for required services dynamically.

Example

- Smartphone searching for nearby Bluetooth speakers.

4. Service Selection

- Suitable service is selected based on requirements such as speed, availability, or location.

Example

- Choosing nearest cloud server for faster response.

5. Service Access

- After selection, communication is established between service provider and consumer.

Example

- Mobile app connecting to online payment gateway.

Technologies Used for Finding Needed Services

1. UPnP (Universal Plug and Play)

- Automatically discovers devices in local networks.

2. Bluetooth SDP

- Discovers nearby Bluetooth services and devices.

3. Jini

- Java-based distributed service discovery framework.

4. Zeroconf

- Provides automatic networking without manual setup.

Real-Time Example

- In a smart home environment, a smartphone automatically discovers nearby smart TVs, printers, speakers, and IoT devices using service discovery mechanisms and connects to the required service without manual configuration.

Interoperability – Notes

Definition

- Interoperability is the ability of different systems, devices, applications, or networks to communicate, exchange data, and work together effectively.

Features

- Supports communication between different platforms.
- Enables data sharing and integration.
- Improves compatibility among systems.

Example

- Android mobile communicating with Windows server through internet services.

Applications

- Cloud computing
- Mobile communication
- IoT systems
- Banking networks

Standardization – Notes

Definition

- Standardization is the process of establishing common rules, protocols, and guidelines to ensure compatibility and uniformity between systems and technologies.

Features

- Ensures compatibility between devices and software.
- Improves communication and reliability.
- Reduces system complexity.

Example

- Wi-Fi standards allowing different devices to connect to the same wireless network.

Applications

- Wireless communication systems
- Internet protocols
- Software development
- Networking technologies

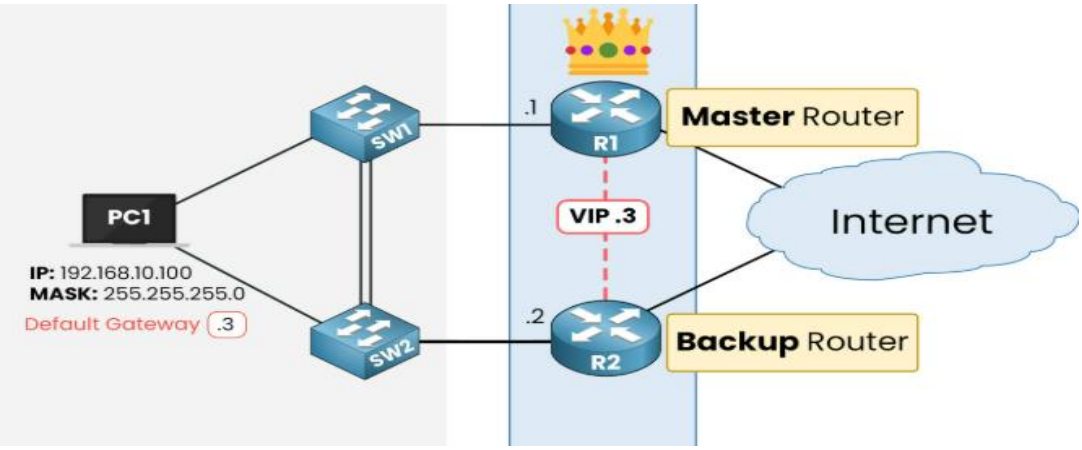
UNIT III – Mobile Networking: Virtual IP Protocols - Loose Source Routing Protocols - Mobile IP – CDPD – GPRS – UMTS - Security and Authentication – Quality of Service – Mobile Access to the World Wide Web

[PAGE NO : 51 TO 74]

Virtual IP Protocols

1. Introduction

- Virtual IP Protocols are network protocols used to provide high availability, fault tolerance, and continuous network connectivity by using a virtual IP address shared among multiple servers or routers.
- A virtual IP address (VIP) is not permanently assigned to one device; instead, it can move between systems when failures occur.
- These protocols are mainly used in load balancing, failover systems, clustering, and network redundancy environments.



Features of Virtual IP Protocols

- Provides high availability of network services.
- Supports automatic failover during server or router failure.
- Improves reliability and fault tolerance.
- Reduces network downtime.
- Supports load balancing and redundancy.

Components of Virtual IP Protocols

1. Virtual IP Address (VIP)

- Shared IP address used by multiple servers or routers.
- Users access services through this address.

Example

- A website accessed using one virtual IP even though multiple servers exist behind it.

2. Primary Server / Router

- Main system handling network traffic normally.

Example

- Active router managing internet communication.

3. Backup Server / Router

- Secondary system that takes control if primary system fails.

Example

- Backup router activated automatically during failure.

Working of Virtual IP Protocols

1. Normal Operation

- Primary server or router handles all traffic using virtual IP address.

2. Monitoring

- Backup systems continuously monitor the active system status.

3. Failure Detection

- If primary system fails, backup system detects failure automatically.

4. Failover Process

- Virtual IP address is transferred to backup system.
- Communication continues without interruption.

Types of Virtual IP Protocols

1. VRRP (Virtual Router Redundancy Protocol)

- Provides automatic backup for routers using virtual IP addresses.

Example

- Backup router taking control when main router fails.

2. HSRP (Hot Standby Router Protocol)

- Cisco protocol used for router redundancy and failover support.

Example

- Enterprise network using standby routers.

3. CARP (Common Address Redundancy Protocol)

- Open-source protocol providing redundancy between servers or routers.

Example

- Linux-based server failover systems.

Applications of Virtual IP Protocols

- Data centers.
- Cloud computing systems.
- Banking and financial networks.
- Enterprise networks.
- High-availability web servers.

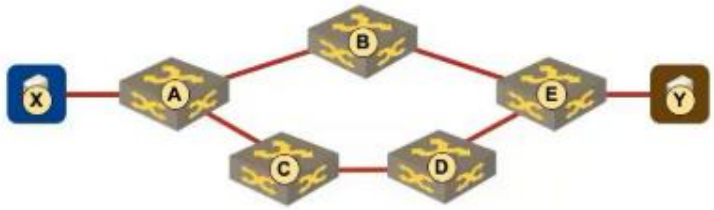
Real-Time Example

- In online banking systems, Virtual IP Protocols ensure continuous service availability by automatically switching traffic to backup servers when the primary server fails, preventing downtime for users.

Loose Source Routing Protocols

1. Introduction

- Loose Source Routing is a routing technique in computer networks where the sender partially specifies the path that packets should follow through the network.
- In this method, the sender provides only selected intermediate routers, while the remaining path is decided dynamically by the network routers.
- It is mainly used for flexible routing, network testing, and troubleshooting.



Concepts

- Source node knows loose path to the destination node
- Loose source route is prepended to every packet sent by the source node
- Intermediate nodes perform hop-by-hop destination-based routing toward next hop in source route

Examples

- Loose source routing in IPv4 or IPv6 (deprecated)
- Traffic engineering or service chaining in segment routing (MPLS or IPv6 variants)

Working of Loose Source Routing

1. Path Selection

- Sender selects important intermediate routers through which packet should travel.

Example

- Packet must pass through Router A and Router C.

2. Packet Transmission

- Packet is transmitted with routing information included in the header.

3. Dynamic Routing

- Routers determine remaining path between specified routers dynamically.

4. Packet Delivery

- Packet reaches destination after passing through selected routers.

Components of Loose Source Routing

1. Source Node

- Device that sends the packet and specifies partial route.

Example

- Client computer sending data packet.

2. Intermediate Routers

- Routers specified by sender or selected dynamically during routing.

Example

- Routers forwarding packet toward destination.

3. Destination Node

- Final receiver of the packet.

Example

- Server receiving network request.

Limitations of Loose Source Routing

- Security risks due to route manipulation.
- Increased routing overhead.
- Not commonly supported in modern secure networks.

Applications of Loose Source Routing

- Network testing and diagnostics.
- Traffic engineering.

- Research and experimental networks.
- Troubleshooting communication paths.

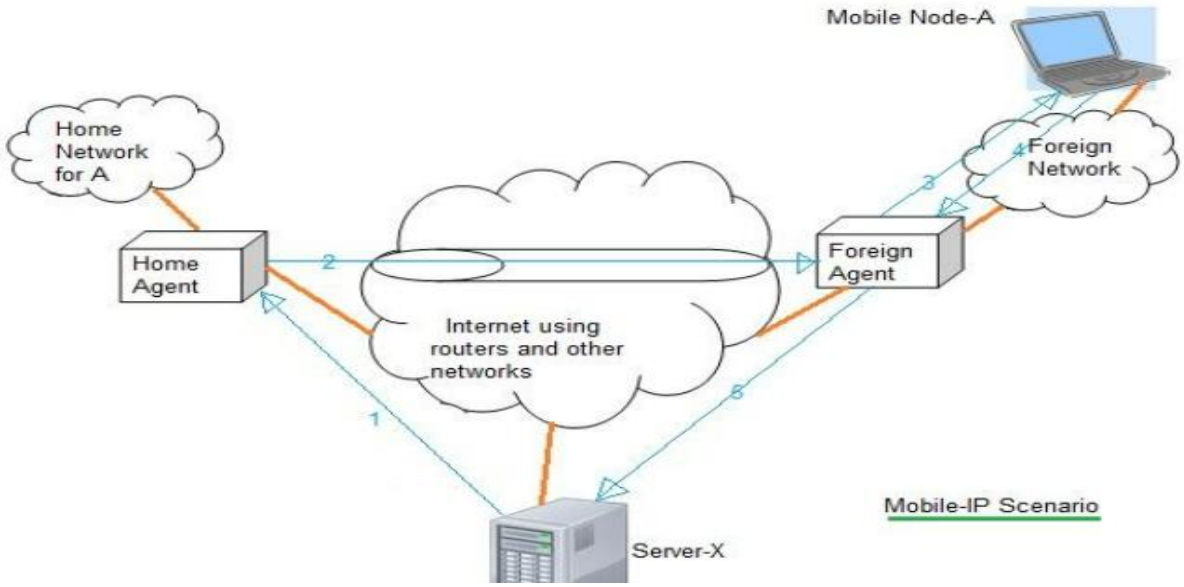
Real-Time Example

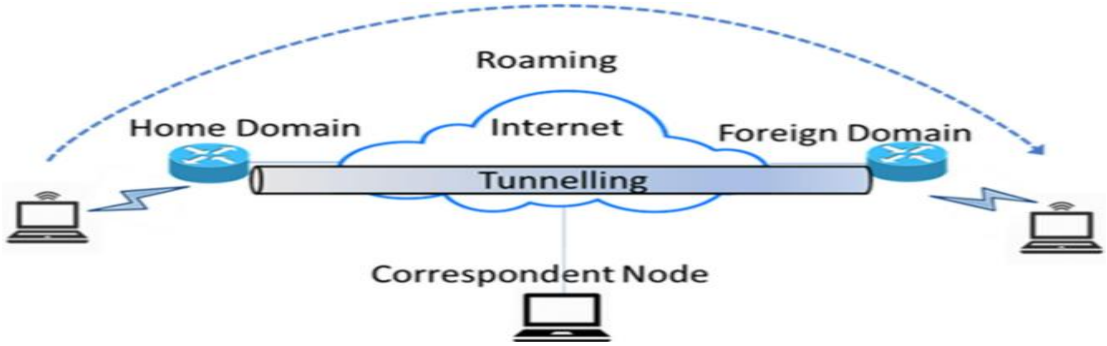
- In a corporate network, administrators may use loose source routing to ensure packets pass through specific monitoring routers before reaching the destination server for traffic analysis and security monitoring.

Mobile IP

1. Introduction

- Mobile IP is a communication protocol that allows mobile devices to move from one network to another while maintaining the same IP address.
- It enables uninterrupted internet connectivity and communication for mobile users.
- Mobile IP is mainly used in wireless and mobile communication networks to support mobility management.
- It allows users to continue ongoing sessions even when changing network locations.





Components of Mobile IP

1. Mobile Node (MN)

- Mobile node is the mobile device that changes its location between networks.

Example

- Smartphone or laptop moving between Wi-Fi networks.

2. Home Agent (HA)

- Home agent is a router located in the home network of the mobile node.
- It maintains current location information of the mobile node.

Functions

- Tracks mobile node location.
- Forwards packets to mobile node.

3. Foreign Agent (FA)

- Foreign agent is a router in the visited network.
- It provides temporary connectivity for mobile nodes.

Functions

- Receives packets from home agent.
- Delivers packets to mobile node.

4. Care-of Address (CoA)

- Temporary address assigned to mobile node in visited network.
- Used to identify current location of the mobile device.

Working of Mobile IP

1. Registration

- Mobile node registers its current location with home agent through foreign agent.

2. Packet Forwarding

- Home agent intercepts packets sent to mobile node.
- Packets are tunneled to foreign agent using care-of address.

3. Packet Delivery

- Foreign agent receives packets and forwards them to mobile node.

4. Mobility Support

- Communication continues even when user moves between networks.

Types of Mobile IP

1. IPv4 Mobile IP

- Supports mobility management in IPv4 networks.

Example

- Traditional wireless communication systems.

2. Mobile IPv6

- Improved version supporting direct routing and enhanced security.

Example

- Modern wireless and IoT systems.

Advantages of Mobile IP

- Provides uninterrupted communication during mobility.
- Supports roaming across networks.
- Maintains same IP address while moving.
- Improves flexibility in wireless communication.

Applications of Mobile IP

- Smartphones and mobile devices.
- Wireless communication systems.
- Mobile banking applications.
- Vehicle communication systems.
- IoT and smart device networks.

Real-Time Example

- When a smartphone user moves from home Wi-Fi to mobile data network during a video call, Mobile IP helps maintain the connection without interrupting the ongoing communication session.

CDPD

1. Introduction

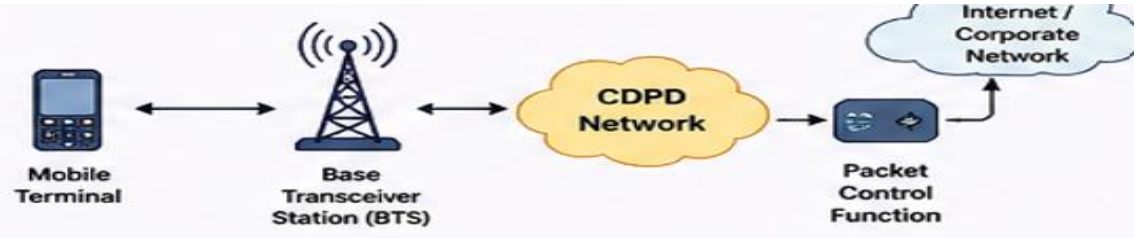
- CDPD stands for Cellular Digital Packet Data.
- It is a wireless communication technology used to transmit packet-switched data over existing cellular networks.
- CDPD allows mobile devices to access internet and data services without interrupting voice communication.
- It was mainly developed for mobile data communication before modern technologies like GPRS and UMTS.

Features of CDPD

- Uses packet-switched communication.
- Supports wireless internet access.
- Operates on existing cellular networks.
- Allows simultaneous voice and data communication.
- Provides mobility support for mobile users.

Working of CDPD

- Data is divided into packets before transmission.
- Packets are transmitted through cellular channels.
- Mobile devices communicate with base stations for internet access.
- Routing and packet management are handled by network infrastructure.



Applications of CDPD

- Mobile internet access.
- Vehicle tracking systems.
- Wireless banking services.
- Mobile email communication.

Real-Time Example

- Delivery vehicles using wireless data communication for location tracking and updates.

GPRS

1. Introduction

- GPRS stands for General Packet Radio Service.
- It is a packet-switched mobile data technology used in GSM networks.
- GPRS provides internet access, multimedia messaging, and wireless application services.
- It is often called 2.5G technology because it improves data communication in 2G networks.

Features of GPRS

- Packet-switched data communication.
- Faster internet speed than GSM.

- Supports multimedia messaging and mobile internet.
- Allows simultaneous voice and data communication.
- Efficient bandwidth usage.

Components of GPRS

1. Mobile Station

- User device such as smartphone or mobile phone.

2. Base Station System

- Handles wireless communication between user and network.

3. SGSN (Serving GPRS Support Node)

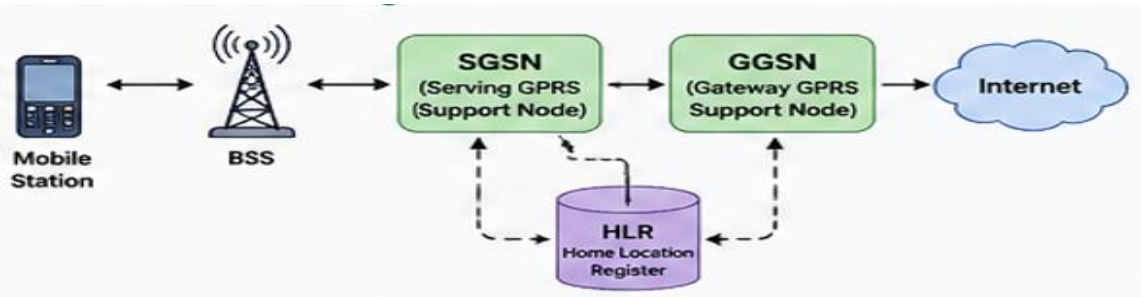
- Manages mobility and packet delivery.

4. GGSN (Gateway GPRS Support Node)

- Connects GPRS network to internet and external networks.

Working of GPRS

- User data is divided into packets.
- Packets are transmitted through GSM network infrastructure.
- SGSN and GGSN manage routing and internet connectivity.



Applications of GPRS

- Mobile internet browsing.
- MMS messaging.
- GPS tracking systems.
- Mobile banking services.

Real-Time Example

- Using internet browsing and WhatsApp messaging on early GSM smartphones.

UMTS

1. Introduction

- UMTS stands for Universal Mobile Telecommunications System.
- It is a 3G wireless communication technology that provides high-speed voice and data services.
- UMTS supports video calling, multimedia services, internet access, and mobile TV.
- It uses WCDMA technology for high-speed wireless communication.

Features of UMTS

- High-speed mobile internet access.
- Supports video calling and multimedia applications.
- Better voice quality and network capacity.
- Global roaming support.
- Improved security and data transmission speed.

Components of UMTS

1. User Equipment (UE)

- Mobile devices such as smartphones and tablets.

2. Node B

- Base station responsible for wireless communication.

3. Radio Network Controller (RNC)

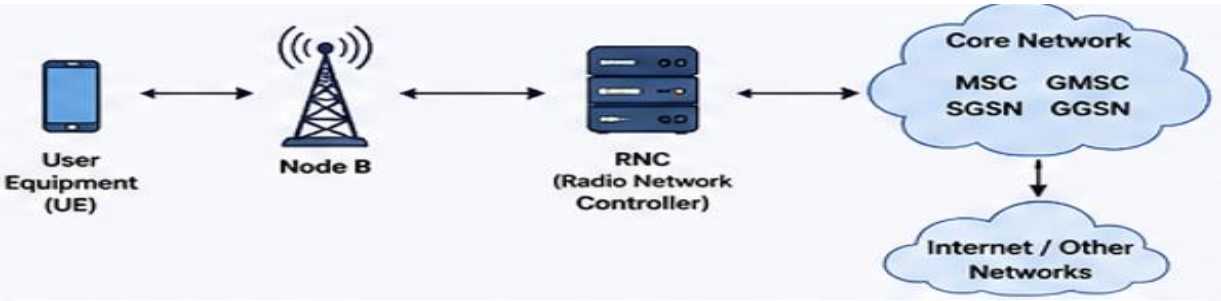
- Controls multiple base stations and manages radio resources.

4. Core Network

- Provides switching, routing, and internet connectivity.

Working of UMTS

- User devices communicate with Node B using WCDMA technology.
- RNC controls communication and mobility management.
- Core network handles internet and voice services.



Applications of UMTS

- Video conferencing.
- Mobile internet browsing.
- Online gaming.
- Mobile TV and multimedia streaming.

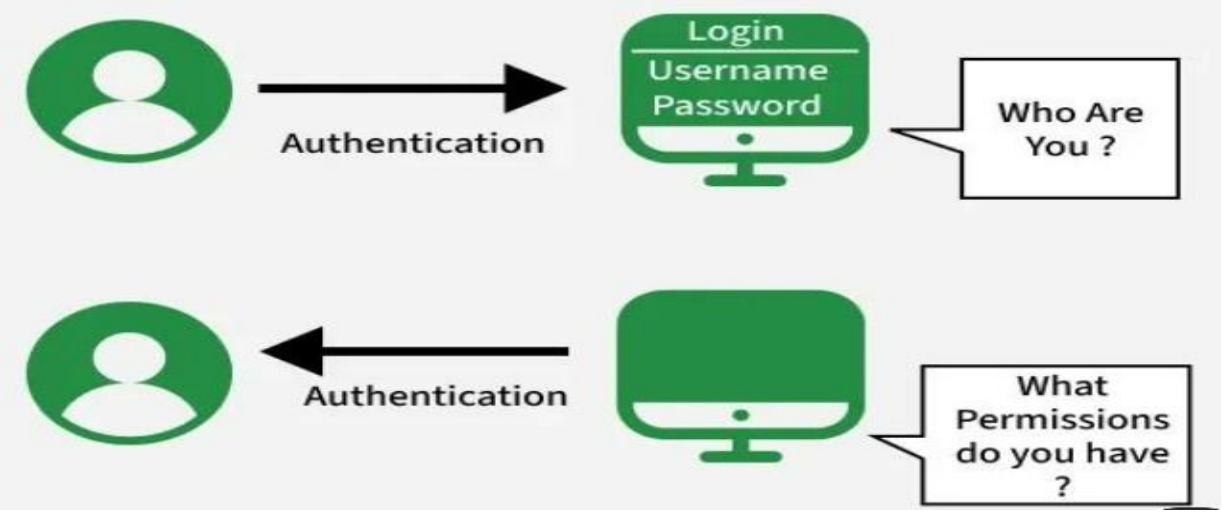
Real-Time Example

- Smartphones using 3G networks for video calling and high-speed internet browsing.

Security and Authentication

1. Introduction

- Security and Authentication are important concepts in computer networks, mobile computing, and information systems.
- Security protects systems, networks, and data from unauthorized access, attacks, and misuse.
- Authentication verifies the identity of users, devices, or applications before granting access to resources.
- Together, they ensure confidentiality, integrity, and availability of information.



Security

Definition

- Security is the process of protecting data, systems, and networks from unauthorized access, modification, destruction, or cyberattacks.

Features of Security

- Protects sensitive information.
- Prevents unauthorized access.
- Ensures secure communication.
- Supports privacy and data integrity.
- Reduces cyber threats and attacks.

Types of Security

1. Network Security

- Protects computer networks from attacks and unauthorized access.

Example

- Firewalls protecting organizational networks.

2. Data Security

- Protects stored and transmitted data.

Example

- Encryption of banking transactions.

3. Application Security

- Protects software applications from vulnerabilities and attacks.

Example

- Secure login systems in web applications.

4. Mobile Security

- Protects smartphones and mobile applications from threats.

Example

- Fingerprint lock in smartphones.

Security Mechanisms

1. Encryption

- Converts data into unreadable format to prevent unauthorized access.

Example

- HTTPS encryption in websites.

2. Firewall

- Monitors and controls incoming and outgoing network traffic.

Example

- Blocking unauthorized network connections.

3. Antivirus Software

- Detects and removes malicious software.

Example

- Windows Defender antivirus protection.

4. Intrusion Detection System (IDS)

- Detects suspicious activities and attacks in networks.

Example

- Detecting hacking attempts in servers.

Authentication

Definition

- Authentication is the process of verifying the identity of users, devices, or systems before allowing access.

Features of Authentication

- Verifies legitimate users.
- Prevents unauthorized access.
- Supports secure communication.
- Protects sensitive information and resources.

Types of Authentication

1. Password-Based Authentication

- Users log in using username and password.

Example

- Email account login.

2. Biometric Authentication

- Uses physical characteristics for verification.

Example

- Fingerprint and face recognition.

3. Token-Based Authentication

- Uses tokens or OTPs for user verification.

Example

- Banking OTP authentication.

4. Multi-Factor Authentication (MFA)

- Uses multiple verification methods for higher security.

Example

- Password + OTP verification in banking apps.

Applications of Security and Authentication

- Online banking systems.
- Mobile payment applications.
- Cloud computing systems.
- E-commerce websites.
- Healthcare information systems.

Real-Time Example

- In online banking applications, security mechanisms like encryption and firewalls protect user data, while authentication methods such as passwords, OTPs, and fingerprint verification ensure only authorized users can access bank accounts.

Quality of Service (QoS)

1. Introduction

- Quality of Service (QoS) refers to the ability of a network to provide better and reliable communication services to users and applications.
- QoS ensures efficient handling of network traffic by controlling bandwidth, delay, packet loss, and data transmission priority.

- It is mainly used in multimedia communication, video conferencing, online gaming, and real-time applications where network performance is very important.



Features of Quality of Service

- Provides reliable network communication.
- Controls network traffic efficiently.
- Reduces delay and packet loss.
- Supports priority-based data transmission.
- Improves performance of real-time applications.

QoS Parameters

1. Bandwidth

- Bandwidth refers to the amount of data that can be transmitted through a network in a given time.

Example

- High-speed internet connection for video streaming.

2. Delay (Latency)

- Delay is the time taken for data to travel from sender to receiver.

Example

- Delay during online video calls.

3. Jitter

- Jitter is the variation in packet arrival time during communication.

Example

- Disturbance in voice calls due to inconsistent packet delivery.

4. Packet Loss

- Packet loss occurs when some data packets fail to reach destination.

Example

- Missing audio during online meetings.

5. Throughput

- Throughput is the actual amount of data successfully transmitted over the network.

Example

- Download speed in internet communication.

QoS Mechanisms

1. Traffic Prioritization

- Important data packets are given higher priority during transmission.

Example

- Voice packets prioritized over email data.

2. Traffic Shaping

- Controls network traffic flow to avoid congestion.

Example

- Limiting bandwidth usage for certain applications.

3. Resource Reservation

- Network resources are reserved for critical applications.

Example

- Reserved bandwidth for video conferencing.

4. Congestion Control

- Prevents network overload and packet loss.

Example

- Managing heavy internet traffic during peak hours.

Types of QoS

1. Best Effort Service

- Network provides service without guarantee of performance.

Example

- Normal web browsing.

2. Integrated Services (IntServ)

- Provides guaranteed QoS using resource reservation.

Example

- Video conferencing systems.

3. Differentiated Services (DiffServ)

- Provides different priority levels for network traffic.

Example

- Prioritizing voice calls over file downloads.

Applications of QoS

- Video conferencing systems.
- Online gaming applications.
- VoIP communication.
- Multimedia streaming services.
- Cloud computing systems.

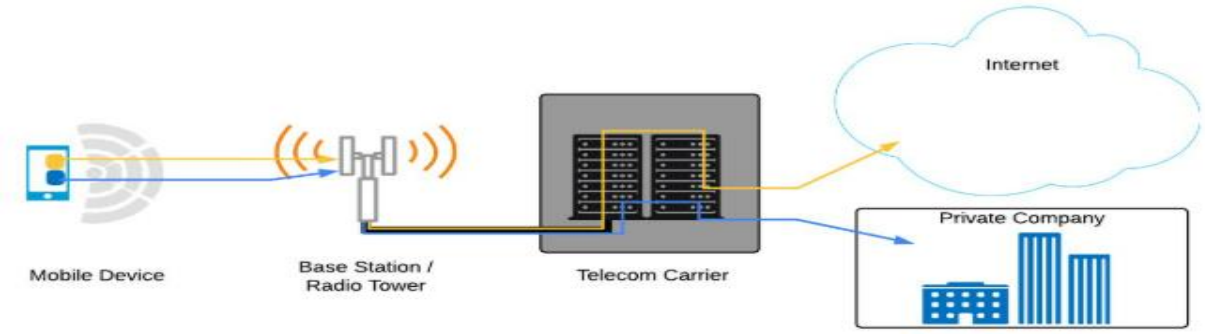
Real-Time Example

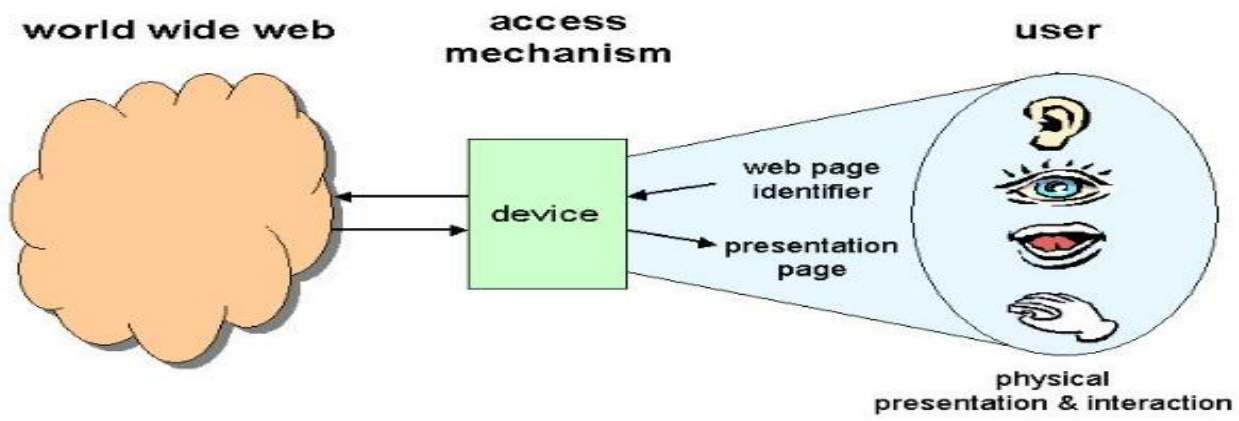
- In online video conferencing applications like Zoom, QoS ensures smooth audio and video communication by reducing delay, minimizing packet loss, and prioritizing real-time voice and video traffic over other network data.

Mobile Access to the World Wide Web

1. Introduction

- Mobile Access to the World Wide Web refers to accessing websites, internet services, and web applications through mobile devices such as smartphones, tablets, and laptops using wireless communication technologies.
- It allows users to browse websites, access online services, communicate, and perform internet-based activities from any location.
- Mobile web access is supported by wireless technologies such as Wi-Fi, 3G, 4G, and 5G networks.





Components of Mobile Web Access

1. Mobile Device

- Devices such as smartphones and tablets used to access the internet.

Example

- Android and iPhone devices.

2. Wireless Network

- Provides internet connectivity through wireless communication technologies.

Example

- Wi-Fi, 4G, and 5G networks.

3. Web Browser

- Software used to access and display web pages.

Example

- Google Chrome, Safari, Firefox.

4. Web Server

- Stores and delivers web pages and online services to users.

Example

- Servers hosting websites and applications.

5. Internet Protocols

- Communication protocols used for web access and data transfer.

Example

- HTTP and HTTPS protocols.

Working of Mobile Access to WWW

1. User Request

- User enters website address or sends request through browser.

2. Wireless Communication

- Mobile device sends request through wireless network to internet.

3. Server Processing

- Web server receives request and processes it.

4. Data Transmission

- Requested web pages and data are transmitted back to mobile device.

5. Web Page Display

- Browser displays the web page on mobile screen.

Technologies Used in Mobile Web Access

1. WAP (Wireless Application Protocol)

- Early technology used for accessing internet on mobile devices.

Example

- Basic internet browsing on old mobile phones.

2. HTML and XHTML

- Used for designing mobile web pages.

Example

- Mobile-friendly websites.

3. Wi-Fi

- Wireless local area network technology for internet access.

Example

- Internet access in homes and offices.

4. Cellular Networks

- Mobile communication technologies such as 3G, 4G, and 5G.

Example

- Internet access through mobile data services.

Applications of Mobile Access to WWW

- Online banking services.
- Social media applications.
- E-commerce and online shopping.
- Video streaming platforms.
- Online education systems.

Real-Time Example

- A smartphone user accessing online banking, social media, YouTube, and shopping websites using 4G or Wi-Fi internet connection is an example of mobile access to the World Wide Web.

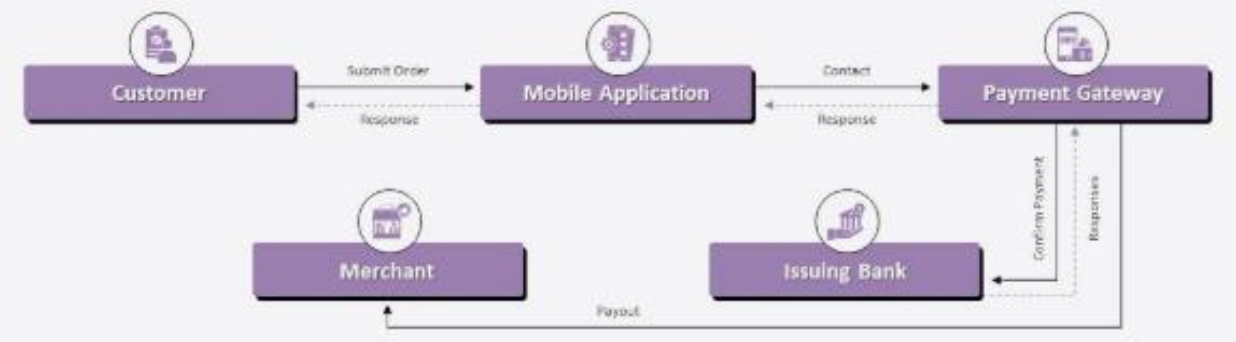
UNIT IV – Mobile Data Management: Mobile Transactions - Reporting and Co Transactions –Kangaroo Transaction Model - Clustering Model –Isolation only transaction – 2 Tier Transaction Model – Semantic based nomadic transaction processing.

[PAGE NO : 75 TO 96]

Mobile Transactions

1. Introduction

- Mobile Transactions refer to electronic transactions performed using mobile devices such as smartphones, tablets, and wireless handheld devices.
- These transactions allow users to perform financial, commercial, and information-related activities through wireless networks and mobile applications.
- Mobile transactions are widely used in banking, online shopping, ticket booking, and digital payment systems.



Components of Mobile Transactions

1. Mobile Device

- Device used by users to perform transactions.

Example

- Smartphones and tablets.

2. Wireless Network

- Provides internet connectivity for transaction communication.

Example

- Wi-Fi, 4G, and 5G networks.

3. Mobile Application

- Software used to perform transactions and access services.

Example

- Google Pay, PhonePe, Paytm applications.

4. Transaction Server

- Processes transaction requests and maintains records.

Example

- Banking transaction servers.

5. Payment Gateway

- Securely transfers payment information between users and banks.

Example

- Razorpay and PayPal payment gateways.

Types of Mobile Transactions

1. Mobile Banking Transactions

- Financial transactions performed using banking mobile applications.

Example

- Fund transfer and balance enquiry.

2. Mobile Payment Transactions

- Digital payments made using smartphones and mobile wallets.

Example

- QR code payment using Google Pay.

3. Mobile Commerce Transactions

- Online shopping and product purchases using mobile devices.

Example

- Purchasing products through Amazon mobile app.

4. Mobile Ticketing Transactions

- Booking travel or movie tickets through mobile applications.

Example

- Train ticket booking using IRCTC app.

Working of Mobile Transactions

1. User Authentication

- User logs into mobile application using password, OTP, or biometric verification.

2. Transaction Request

- User sends request for payment or service through mobile app.

3. Server Processing

- Transaction server verifies and processes the request.

4. Payment Authorization

- Payment gateway and bank verify transaction details.

5. Transaction Completion

- Amount is transferred and confirmation message is sent to user.

Security in Mobile Transactions

1. Encryption

- Protects transaction data during communication.

2. Authentication

- Verifies user identity using passwords, OTPs, or biometrics.

3. Secure Payment Gateway

- Ensures safe transfer of payment information.

4. Digital Signatures

- Used for transaction verification and integrity.

Real-Time Example

- A user paying electricity bill using Google Pay through smartphone connected to 4G network is an example of a mobile transaction.

Reporting and Co-Transactions

1. Introduction

- Reporting and Co-Transactions are important concepts in mobile commerce and distributed transaction systems.
- Reporting refers to the process of collecting, analyzing, and presenting transaction information in the form of reports.
- Co-Transactions refer to coordinated or related transactions that occur together across multiple systems or applications.
- These concepts help organizations monitor activities, maintain records, and ensure consistency in transaction processing.



Reporting

Definition

- Reporting is the process of generating structured information about transactions, activities, or system operations for monitoring and decision-making purposes.

Features of Reporting

- Provides transaction details and summaries.
- Helps in monitoring business activities.
- Supports analysis and decision making.
- Maintains records of operations and transactions.

Types of Reports

1. Transaction Reports

- Contains details of completed transactions.

Example

- Bank account statement showing money transfers.

2. Summary Reports

- Provides summarized information about activities and operations.

Example

- Daily sales report in e-commerce applications.

3. Exception Reports

- Displays unusual or failed transactions.

Example

- Failed online payment report.

4. Audit Reports

- Maintains records for security and verification purposes.

Example

- Banking audit transaction logs.

Functions of Reporting

1. Monitoring

- Helps organizations track system and transaction activities.

2. Analysis

- Supports analysis of performance and operations.

3. Decision Making

- Helps management make business decisions.

4. Record Maintenance

- Maintains historical transaction information.

Co-Transactions

Definition

- Co-Transactions are multiple related transactions that work together and must be coordinated to maintain consistency and reliability.

Types of Co-Transactions

1. Distributed Transactions

- Transactions executed across multiple networked systems.

Example

- Online banking transaction involving different bank servers.

2. Cooperative Transactions

- Multiple systems cooperate to complete a task.

Example

- Airline ticket booking with payment gateway integration.

3. Concurrent Transactions

- Multiple transactions occurring simultaneously.

Example

- Multiple users purchasing products in e-commerce systems.

Working of Co-Transactions

1. Transaction Initiation

- User or system starts multiple related transactions.

2. Coordination

- Transaction manager coordinates operations between systems.

3. Processing

- Systems process requests simultaneously or sequentially.

4. Synchronization

- Data consistency and integrity are maintained.

5. Completion

- All related transactions complete successfully together.

Applications of Reporting and Co-Transactions

- Banking systems.
- E-commerce platforms.
- Airline reservation systems.
- Mobile payment applications.
- Enterprise resource planning systems.

Real-Time Example

- In online banking, transaction reports provide details of account transfers, while co-transactions coordinate payment processing, account balance updates, and confirmation messages across multiple banking servers simultaneously.

Kangaroo Transaction Model

1. Introduction

- Kangaroo Transaction Model is a mobile transaction model designed for mobile computing environments where users frequently move between different network cells or locations.
- It supports continuous transaction processing even when mobile users change their location during communication.
- The model is called “Kangaroo” because transactions move from one cell to another similar to the jumping movement of a kangaroo.
- It is mainly used in mobile databases and distributed mobile systems.

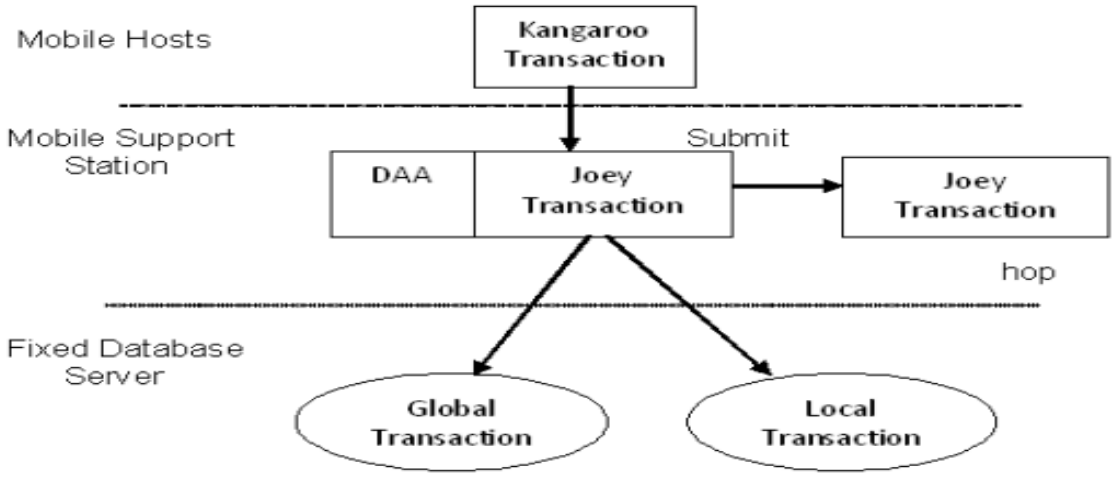


Figure 1- Kangaroo transaction model

Components of Kangaroo Transaction Model

1. Mobile Host (MH)

- Mobile device used by the user to initiate transactions.

Example

- Smartphone or tablet performing mobile banking transaction.

2. Mobile Support Station (MSS)

- Fixed network station that supports communication with mobile hosts.
- Manages transaction processing in a particular network cell.

Example

- Cellular base station handling mobile communication.

3. Kangaroo Transaction

- Main transaction that moves between different cells during user mobility.

Example

- Online ticket booking while user moves between locations.

4. Joey Transactions

- Sub-transactions created within each cell visited by the mobile user.
- Each Joey transaction performs local transaction processing.

Example

- Local processing of payment request in a specific network area.

Working of Kangaroo Transaction Model

1. Transaction Initiation

- User starts transaction using mobile device.

2. Cell-Based Processing

- Transaction is processed within current network cell through MSS.

3. Mobility Support

- When user moves to another cell, transaction control shifts to new MSS.

4. Joey Transaction Creation

- A new Joey transaction is created in the new cell for local processing.

5. Transaction Completion

- All Joey transactions combine to complete the main Kangaroo transaction successfully.

Limitations of Kangaroo Transaction Model

- Complex transaction coordination.
- Increased communication overhead.
- Difficult synchronization between cells.

Applications of Kangaroo Transaction Model

- Mobile banking systems.
- Mobile ticket booking systems.

- Wireless e-commerce applications.
- Vehicle communication systems.
- Distributed mobile databases.

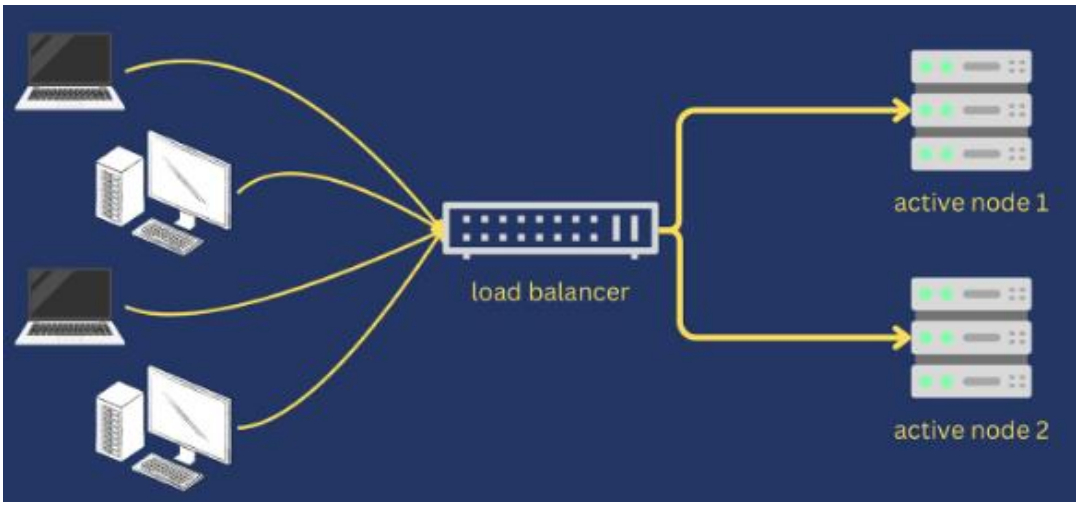
Real-Time Example

- A user booking railway tickets while traveling in a moving vehicle can continue the transaction without interruption even when switching between different mobile network cells using the Kangaroo Transaction Model.

Clustering Model

1. Introduction

- Clustering Model is a technique used in distributed systems, mobile computing, databases, and data analysis to group similar objects, systems, or nodes together into clusters.
- A cluster is a collection of interconnected systems or data elements that work together as a single unit.
- Clustering improves performance, scalability, reliability, and resource management in network and computing environments.



Components of Clustering Model

1. Cluster Nodes

- Individual systems or devices connected together in a cluster.

Example

- Multiple servers working together in a data center.

2. Cluster Manager

- Controls and manages communication between cluster nodes.

Functions

- Load balancing
- Resource allocation
- Failure management

3. Shared Resources

- Common resources shared among cluster nodes.

Example

- Shared databases and storage systems.

4. Communication Network

- Network connecting all cluster nodes for coordination and data exchange.

Example

- High-speed LAN in server clusters.

Types of Clustering Models

1. High Availability Clustering

- Ensures continuous system operation even if one node fails.

Example

- Banking servers with backup systems.

2. Load Balancing Clustering

- Distributes workload equally among multiple systems.

Example

- Web servers handling large website traffic.

3. Parallel Clustering

- Multiple nodes process tasks simultaneously for faster execution.

Example

- Scientific computing systems.

4. Database Clustering

- Multiple database servers work together to improve performance and reliability.

Example

- Cloud database systems.

Working of Clustering Model

1. Cluster Formation

- Multiple systems or nodes are connected together.

2. Resource Sharing

- Nodes share resources and workloads.

3. Load Distribution

- Tasks are distributed among cluster nodes.

4. Failure Handling

- If one node fails, another node continues processing.

5. Communication and Synchronization

- Nodes communicate continuously to maintain consistency.

Limitations of Clustering Model

- Complex cluster management.
- Expensive infrastructure setup.
- Requires synchronization between nodes.

Applications of Clustering Model

- Cloud computing systems.
- Data centers and web servers.
- Banking and financial systems.
- Distributed databases.
- Scientific and research computing.

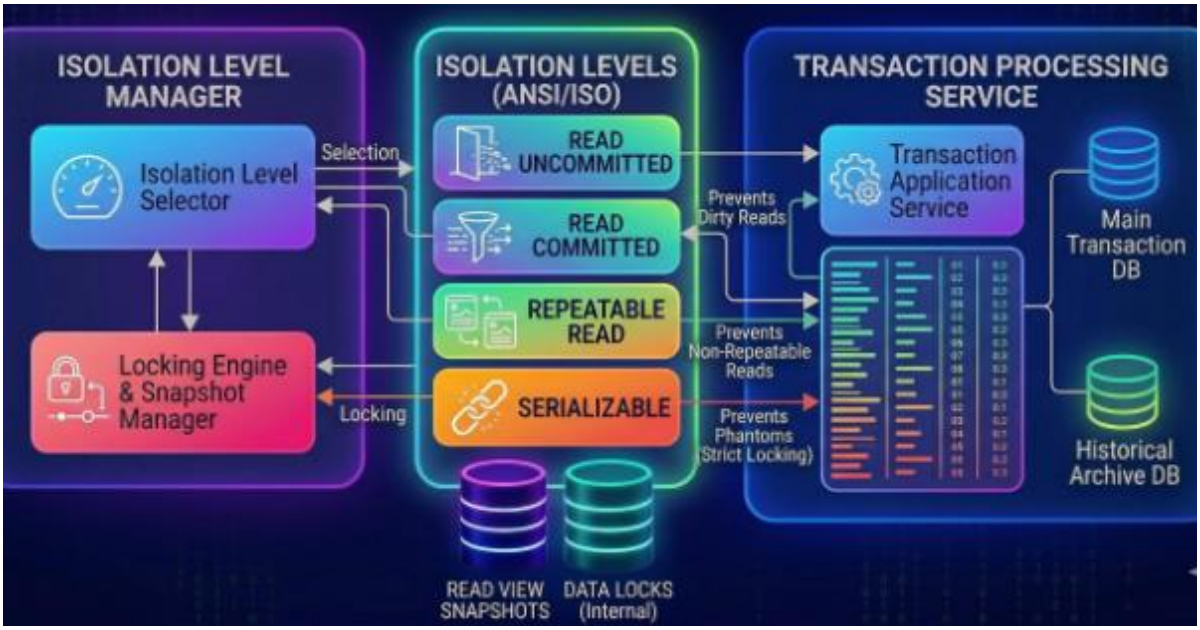
Real-Time Example

- In cloud computing environments, multiple servers are grouped into clusters to distribute workload, provide continuous service, and ensure high availability even if one server fails.

Isolation Only Transaction

1. Introduction

- Isolation Only Transaction is a transaction model used in mobile and distributed database systems.
- It focuses mainly on maintaining isolation between transactions while allowing mobile users to perform operations independently.
- In this model, transactions are executed separately without affecting other concurrent transactions.
- It ensures that intermediate changes made by one transaction are not visible to other transactions until completion.



Components of Isolation Only Transaction

1. Transaction Manager

- Controls execution and coordination of transactions.

Functions

- Manages transaction isolation.
- Controls concurrent access to data.

2. Mobile Host

- Device where transaction is initiated by user.

Example

- Smartphone performing mobile banking operation.

3. Database System

- Stores and manages transaction data securely.

Example

- Banking database server.

4. Concurrency Control Mechanism

- Ensures multiple transactions execute safely without conflicts.

Example

- Locking mechanisms in databases.

Working of Isolation Only Transaction

1. Transaction Initiation

- User starts transaction through mobile or distributed system.

2. Isolation Enforcement

- Transaction executes independently without exposing temporary changes to others.

3. Concurrent Execution

- Multiple transactions may run simultaneously but remain isolated.

4. Transaction Completion

- Changes become visible only after successful commit.

5. Rollback Handling

- If failure occurs, transaction changes are undone without affecting others.

Limitations of Isolation Only Transaction

- Increased transaction management complexity.
- May reduce system performance due to locking.
- High synchronization overhead in distributed systems.

Applications of Isolation Only Transaction

- Mobile banking systems.
- Distributed database systems.
- Airline reservation systems.

- E-commerce transaction processing.
- Cloud computing applications.

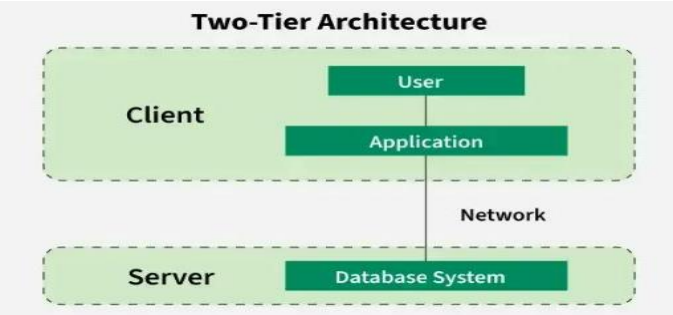
Real-Time Example

- In online banking systems, when two users access the same account simultaneously, Isolation Only Transaction ensures one transaction does not interfere with the other until processing is completed successfully.

2-Tier Transaction Model

1. Introduction

- The 2-Tier Transaction Model is a transaction processing model used in mobile and distributed database systems.
- It divides transaction execution into two levels or tiers to support mobile users and improve transaction management.
- One tier handles mobile operations, while the other tier manages fixed network or database operations.
- This model helps maintain consistency and reliability in mobile computing environments.



Components of 2-Tier Transaction Model

1. Mobile Tier

- Handles transaction processing at the mobile device side.
- Allows users to perform local operations even during weak connectivity.

Example

- Smartphone performing local banking transaction operations.

2. Server Tier

- Handles database management and centralized transaction processing.
- Ensures consistency and synchronization of data.

Example

- Banking server processing account updates.

3. Transaction Manager

- Coordinates communication between mobile tier and server tier.
- Maintains transaction consistency and synchronization.

Functions

- Commit management
- Rollback management
- Concurrency control

Working of 2-Tier Transaction Model

1. Transaction Initiation

- User starts transaction using mobile device.

2. Local Processing

- Mobile tier performs local transaction operations temporarily.

3. Communication with Server

- Mobile tier sends transaction updates to server tier when network is available.

4. Centralized Processing

- Server tier verifies and processes transaction requests.

5. Commit or Rollback

- Transaction manager commits successful transactions or rolls back failed transactions.

Types of Transactions in 2-Tier Model

1. Local Transactions

- Executed only at mobile device level.

Example

- Temporary offline data entry.

2. Global Transactions

- Involves communication with central database server.

Example

- Online bank money transfer.

Limitations of 2-Tier Transaction Model

- Complex synchronization between tiers.
- Network failures may delay updates.
- Increased transaction management overhead.

Applications of 2-Tier Transaction Model

- Mobile banking systems.
- Mobile ticket booking systems.
- Distributed database systems.
- Wireless e-commerce applications.
- Cloud-based mobile applications.

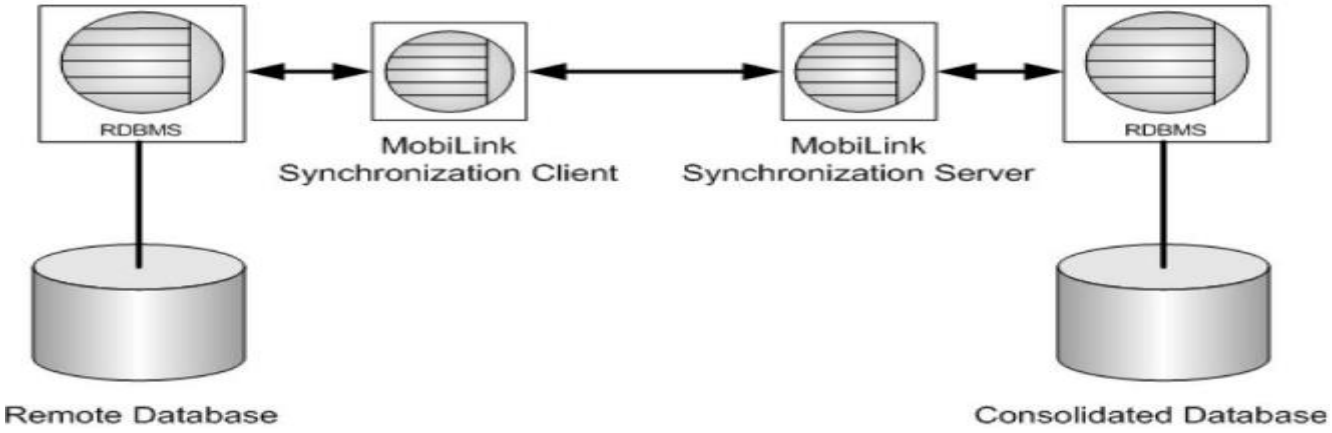
Real-Time Example

- In mobile banking applications, users can perform local transaction operations on smartphones, and when internet connectivity becomes available, transaction details are synchronized with central banking servers using the 2-Tier Transaction Model.

Semantic-Based Nomadic Transaction Processing

1. Introduction

- Semantic-Based Nomadic Transaction Processing is a transaction management approach used in mobile and distributed computing environments.
- It supports mobile users who frequently move between different locations and networks while performing transactions.
- The model uses semantic information about transactions to improve flexibility, consistency, and transaction execution during mobility.
- It allows transactions to continue even when network disconnections or mobility changes occur.



Components of Semantic-Based Nomadic Transaction Processing

1. Mobile User

- User who performs transactions while moving between different locations.

Example

- Smartphone user performing online banking while traveling.

2. Mobile Device

- Device used to initiate and manage transactions.

Example

- Smartphone, tablet, or laptop.

3. Transaction Manager

- Controls execution, synchronization, and consistency of transactions.

Functions

- Transaction coordination
- Commit and rollback management
- Synchronization handling

4. Semantic Information

- Describes meaning and relationships of transaction operations.
- Helps system decide how transactions can be processed during mobility.

Example

- Identifying which transaction operations are independent or dependent.

5. Database Server

- Stores and manages transaction data centrally.

Example

- Banking database server.

Working of Semantic-Based Nomadic Transaction Processing

1. Transaction Initiation

- User starts transaction through mobile device.

2. Semantic Analysis

- System analyzes transaction semantics and dependencies.

3. Local Processing

- Some transaction operations are processed locally during mobility or disconnection.

4. Synchronization

- When network becomes available, local changes synchronize with central database.

5. Commit or Rollback

- Transaction manager finalizes transaction after verification.

Limitations of Semantic-Based Nomadic Transaction Processing

- Complex semantic analysis and management.
- Difficult synchronization in distributed environments.
- Increased transaction processing overhead.

Applications of Semantic-Based Nomadic Transaction Processing

- Mobile banking systems.
- Wireless e-commerce applications.
- Mobile healthcare systems.
- Distributed mobile databases.
- Smart transportation systems.

Real-Time Example

- A traveler using mobile banking during weak network connectivity can continue transaction operations locally, and later synchronize updates with the banking server when internet connectivity becomes available using Semantic-Based Nomadic Transaction Processing.

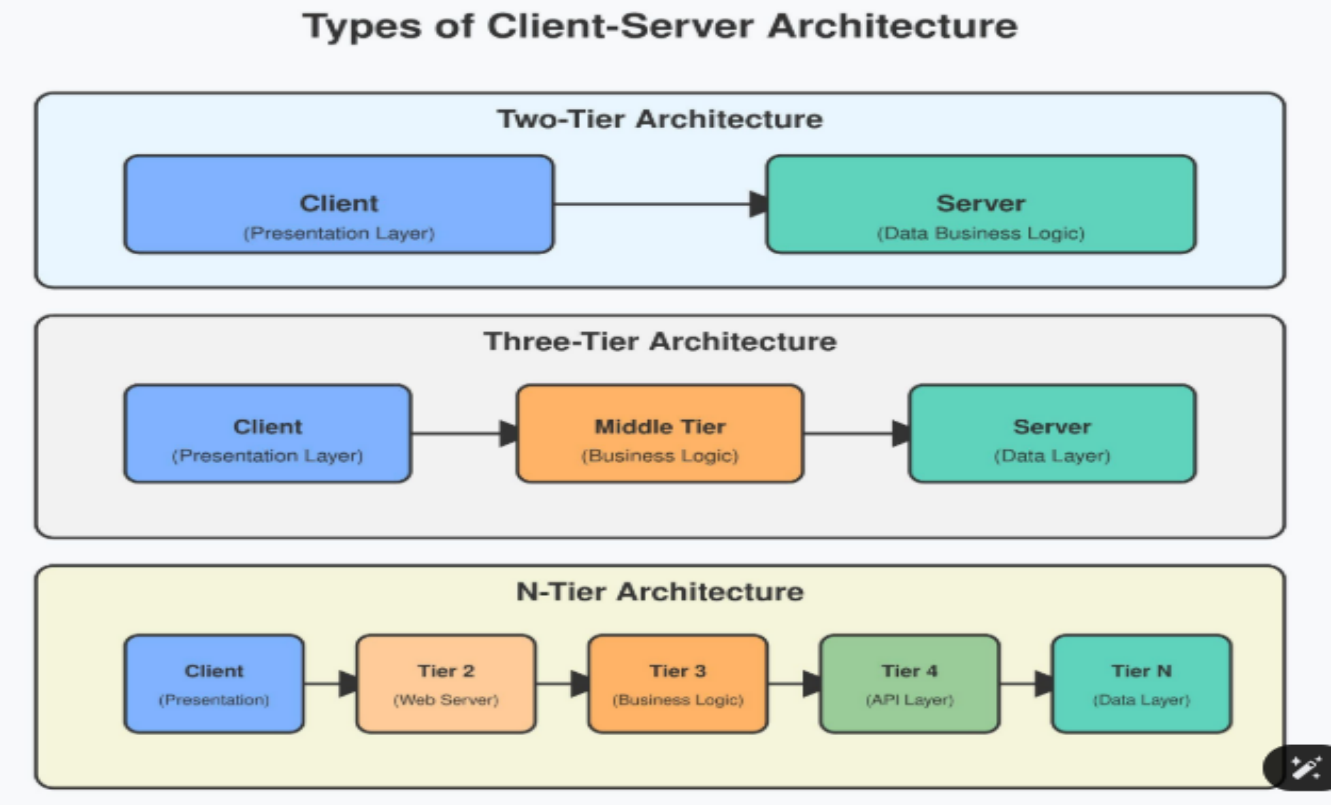
UNIT V – Mobile Computing Models: Client Server model – Client/Proxy/Server Model – Disconnected Operation Model – Mobile Agent Model – Thin Client Model – Tools: Java, Brew, Windows CE, WAP, Sybian, and EPOC.

[PAGE NO : 97 TO 122]

Client Server Model

1. Introduction

- Client Server Model is a network architecture in which multiple clients request services or resources from a centralized server.
- The client is the user device or application that sends requests, while the server processes requests and provides services or data.
- It is one of the most widely used models in networking, web applications, databases, and distributed systems.





Components of Client Server Model

1. Client

- The client is a user device or application that requests services from the server.

Example

- Web browser, mobile app, or desktop application.

2. Server

- The server is a powerful system that provides services, resources, and data to clients.

Functions

- Data processing
- Request handling
- Resource management

Example

- Web server or database server.

3. Network

- Communication medium connecting clients and servers.

Example

- Internet, LAN, Wi-Fi networks.

4. Communication Protocol

- Rules and standards used for communication between client and server.

Example

- HTTP, HTTPS, TCP/IP protocols.

Working of Client Server Model

1. Client Request

- Client sends request to server through network.

Example

- User requesting webpage through browser.

2. Server Processing

- Server receives request and processes it.

Example

- Web server retrieving requested webpage data.

3. Response Generation

- Server prepares response after processing request.

Example

- Server generating webpage output.

4. Response Delivery

- Server sends response back to client.

Example

- Webpage displayed in browser.

Types of Client Server Models

1. Two-Tier Architecture

- Client communicates directly with server.

Example

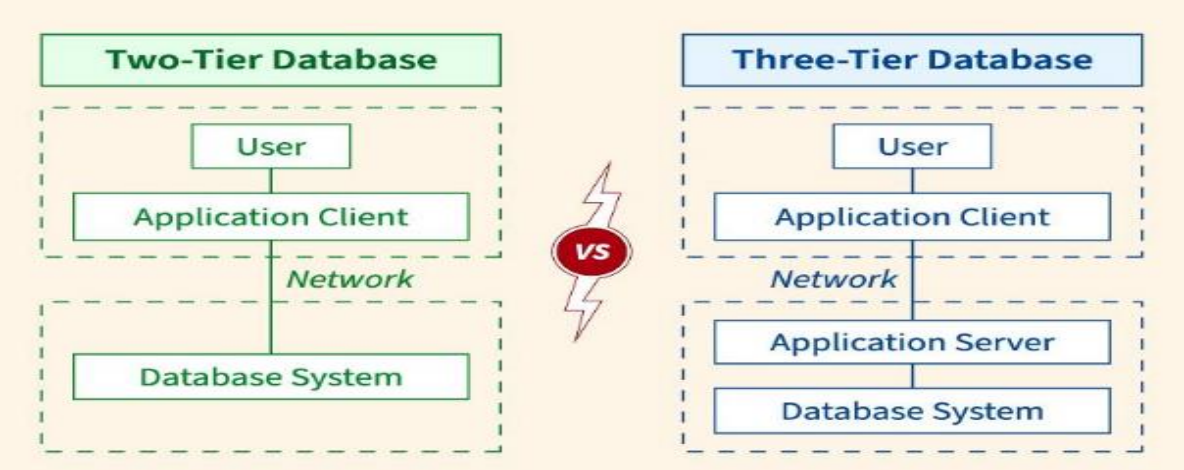
- Desktop application connected to database server.

2. Three-Tier Architecture

- Includes presentation layer, application layer, and database layer.

Example

- Web applications using browser, application server, and database server.



3. N-Tier Architecture

- Uses multiple layers for scalability and distributed processing.

Example

- Large cloud computing systems.

Limitations of Client Server Model

- Server failure affects all clients.
- High network dependency.

- Requires powerful server infrastructure.
- Heavy traffic may reduce performance.

Applications of Client Server Model

- Web applications.
- Online banking systems.
- E-commerce platforms.
- Email services.
- Cloud computing systems.

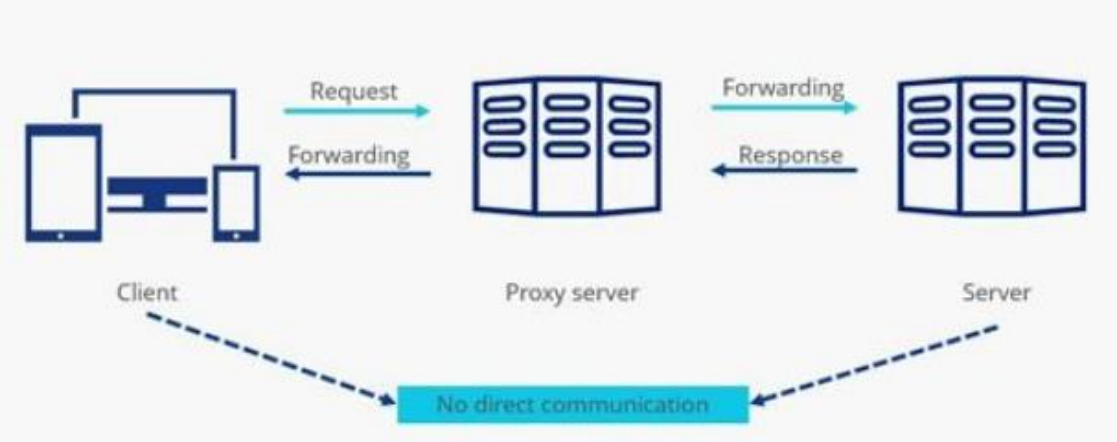
Real-Time Example

- In online banking systems, the mobile banking app acts as the client, while the banking server processes account requests, transactions, and data retrieval through internet communication using the Client Server Model.

Client/Proxy/Server Model

1. Introduction

- Client/Proxy/Server Model is an extended client-server architecture used in distributed systems, mobile computing, and web applications.
- In this model, a proxy server acts as an intermediate layer between client and main server.
- The proxy receives requests from clients, processes or forwards them to the server, and returns responses back to clients.
- This model improves performance, security, caching, and communication efficiency.



Components of Client/Proxy/Server Model

1. Client

- Client is the user device or application requesting services or data.

Example

- Web browser or mobile application.

2. Proxy Server

- Proxy acts as an intermediary between client and main server.
- It receives client requests, processes them, and forwards them to the server if necessary.

Functions

- Request filtering
- Caching
- Security management
- Load balancing

Example

- Proxy server storing frequently accessed web pages.

3. Main Server

- Server provides requested services, resources, or data.

Functions

- Request processing
- Database access
- Service management

Example

- Web server hosting websites and applications.

4. Communication Network

- Network connecting clients, proxy servers, and main servers.

Example

- Internet or LAN network.

Working of Client/Proxy/Server Model

1. Client Request

- Client sends request to proxy server instead of directly contacting main server.

Example

- User requesting webpage through browser.

2. Proxy Processing

- Proxy checks whether requested data is available in cache.
- If available, proxy sends data directly to client.

3. Request Forwarding

- If data is not available, proxy forwards request to main server.

4. Server Response

- Main server processes request and sends response to proxy server.

5. Response Delivery

- Proxy server sends response back to client and may store it for future use.

Types of Proxy Servers

1. Forward Proxy

- Represents clients and forwards requests to servers.

Example

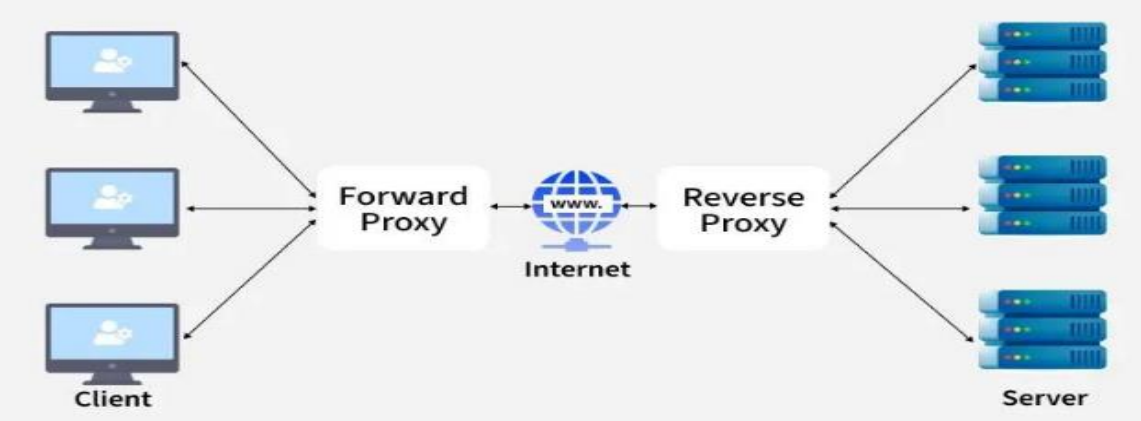
- Proxy used in organizations for internet access control.

2. Reverse Proxy

- Represents servers and handles client requests before forwarding them to backend servers.

Example

- Load balancing web applications.



3. Transparent Proxy

- Operates without requiring client configuration.

Example

- Network monitoring systems.

Limitations of Client/Proxy/Server Model

- Proxy failure affects communication.
- Additional configuration and management required.
- May introduce communication delay in some cases.

Applications of Client/Proxy/Server Model

- Web browsing systems.
- Mobile computing applications.
- Cloud computing systems.

- Enterprise networks.
- Content delivery networks (CDN).

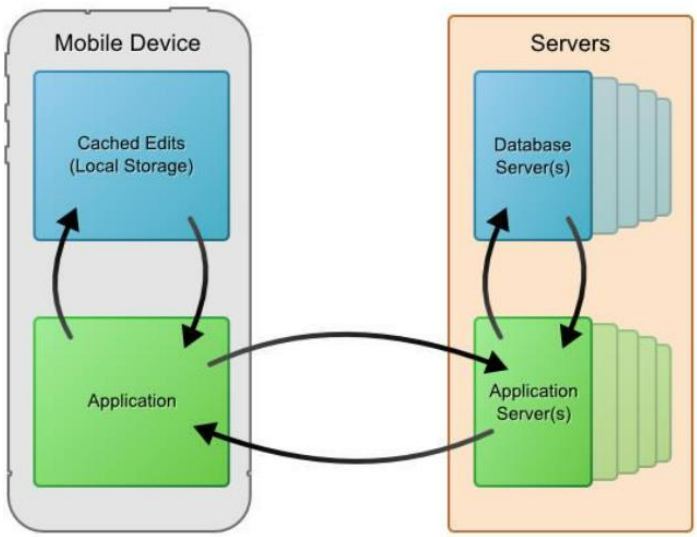
Real-Time Example

- In a company network, employees access websites through a proxy server that filters content, stores frequently accessed web pages in cache, and forwards requests to internet servers securely using the Client/Proxy/Server Model.

Disconnected Operation Model

1. Introduction

- Disconnected Operation Model is a mobile computing model that allows mobile devices to continue working even when network connectivity is temporarily unavailable.
- It supports mobile users who frequently experience weak signals, network failures, or mobility-related disconnections.
- In this model, operations are performed locally on the mobile device and later synchronized with the central server when connection becomes available.
- It improves flexibility and reliability in mobile and distributed systems.



Components of Disconnected Operation Model

1. Mobile Device

- Device used by mobile users for performing operations locally.

Example

- Smartphone, tablet, or laptop.

2. Local Database

- Stores temporary data and transaction information on mobile device.

Example

- Offline storage in mobile banking apps.

3. Synchronization Manager

- Synchronizes local updates with central server after reconnection.

Functions

- Data synchronization
- Conflict resolution
- Update management

4. Central Server

- Stores main database and manages global system operations.

Example

- Banking or cloud server.

Working of Disconnected Operation Model

1. Normal Connected Operation

- Mobile device communicates with central server through network.

2. Disconnection

- Network becomes unavailable due to mobility or signal loss.

3. Local Processing

- User continues operations locally using cached or stored data.

Example

- Writing emails offline in mobile applications.

4. Data Storage

- Changes and transactions are stored temporarily in local database.

5. Reconnection and Synchronization

- When network becomes available, local updates synchronize with central server.

6. Conflict Resolution

- Any data conflicts between local and server updates are resolved.

Limitations of Disconnected Operation Model

- Data synchronization complexity.
- Possible data inconsistency conflicts.
- Increased local storage requirement.

Applications of Disconnected Operation Model

- Mobile banking applications.
- Offline email systems.
- Mobile healthcare systems.
- E-learning applications.
- Cloud-based mobile applications.

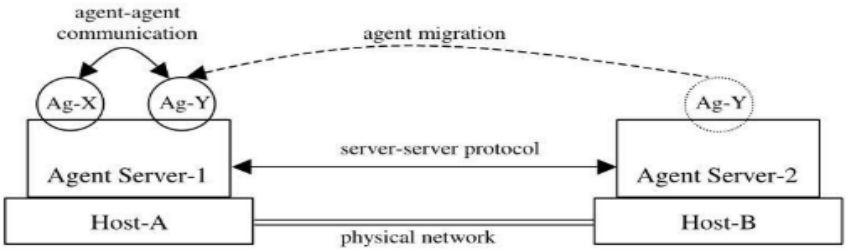
Real-Time Example

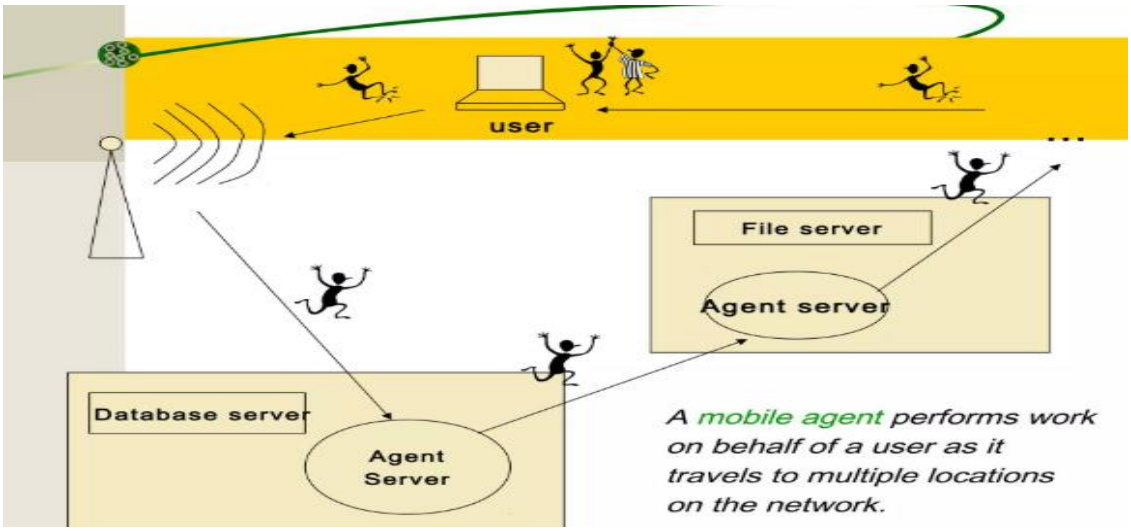
- In Google Docs or offline email applications, users can continue editing documents or writing emails without internet connection, and all updates are synchronized automatically when connectivity is restored using the Disconnected Operation Model.

Mobile Agent Model

1. Introduction

- Mobile Agent Model is a distributed computing model in which software agents move from one computer or network to another to perform tasks automatically.
- A mobile agent is an intelligent software program that carries data, code, and execution state while moving across networks.
- Mobile agents reduce network traffic and improve flexibility in distributed and mobile computing environments.
- This model is widely used in mobile computing, distributed systems, e-commerce, and network management.





Components of Mobile Agent Model

1. Mobile Agent

- Software entity that moves across systems to perform tasks.

Functions

- Data collection
- Task execution
- Decision making

Example

- Shopping agent collecting product prices from different websites.

2. Agent Platform

- Environment where mobile agents execute and communicate.

Functions

- Agent execution
- Resource management
- Security handling

Example

- Java-based mobile agent platforms.

3. Host System

- Computer or server where mobile agents execute tasks.

Example

- Web server processing e-commerce requests.

4. Communication Network

- Network connecting hosts and supporting agent movement.

Example

- Internet or wireless network.

Working of Mobile Agent Model

1. Agent Creation

- Mobile agent is created with required code and task information.

2. Agent Migration

- Agent moves from one host system to another through network.

3. Task Execution

- Agent performs operations at destination system locally.

Example

- Collecting information from remote database.

4. Result Collection

- Agent gathers results after processing.

5. Return or Further Migration

- Agent returns results to user or moves to another host for additional tasks.

Types of Mobile Agents

1. Collaborative Agents

- Work together with other agents to complete tasks.

Example

- Multi-agent robotic systems.

2. Information Agents

- Collect and manage information from multiple sources.

Example

- Search engine data collection agents.

3. Transaction Agents

- Handle mobile and distributed transactions.

Example

- Mobile banking transaction processing agents.

4. Monitoring Agents

- Monitor systems and networks continuously.

Example

- Network performance monitoring agents.

Applications of Mobile Agent Model

- E-commerce systems.
- Network management systems.
- Mobile banking applications.
- Information retrieval systems.
- Distributed database systems.

Real-Time Example

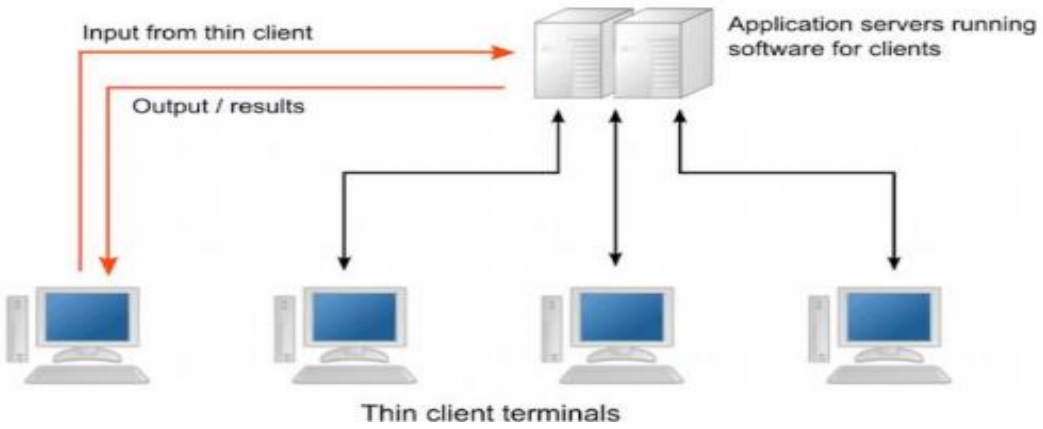
- In e-commerce systems, a mobile agent can travel between multiple online shopping servers, compare product prices, collect details, and return the best result to the user automatically without continuous network communication.

Thin Client Model

1. Introduction

- Thin Client Model is a client-server computing architecture in which the client device performs minimal processing while most processing and data management are handled by the central server.
- Thin clients mainly depend on servers for applications, storage, and processing operations.

- This model is widely used in cloud computing, enterprise networks, web applications, and distributed systems.
- Thin clients are lightweight devices with limited hardware and software resources.



Components of Thin Client Model

1. Thin Client

- Lightweight client device used by end users.
- Mainly handles user interface and input/output operations.

Example

- Low-cost desktop terminal or web browser.

2. Application Server

- Central server that processes applications and user requests.

Functions

- Application execution
- Data processing
- Resource management

Example

- Cloud server hosting enterprise applications.

3. Database Server

- Stores and manages centralized data.

Example

- Banking database server.

4. Communication Network

- Connects thin clients with central servers.

Example

- Internet, LAN, or wireless network.

Working of Thin Client Model

1. User Request

- User interacts with thin client device and sends request.

2. Request Transmission

- Thin client forwards request to central server through network.

3. Server Processing

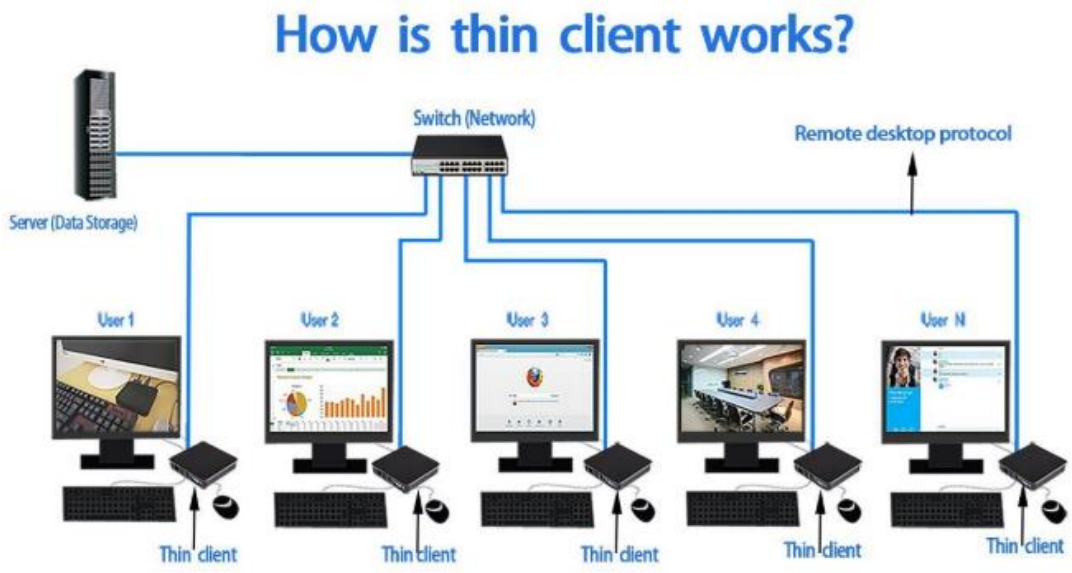
- Server processes application logic and data operations.

4. Response Generation

- Server generates output after processing request.

5. Result Display

- Thin client receives and displays result to user.



Types of Thin Clients

1. Web-Based Thin Client

- Uses web browser for accessing server applications.

Example

- Gmail accessed through browser.

2. Hardware Thin Client

- Dedicated lightweight devices connected to server systems.

Example

- Thin terminals used in offices and labs.

3. Cloud Thin Client

- Accesses cloud-hosted applications and services.

Example

- Google Workspace applications.

Limitations of Thin Client Model

- High dependency on server and network.
- Server failure affects all clients.
- Network delay may reduce performance.
- Requires reliable internet connectivity.

Applications of Thin Client Model

- Cloud computing systems.
- Educational computer labs.
- Banking and enterprise systems.
- Web applications.
- Virtual desktop environments.

Real-Time Example

- In schools and offices, thin client computers are connected to a central server where all applications and data are processed, while users access software and services through lightweight client terminals using the Thin Client Model.

TOOLS :

Java

Definition

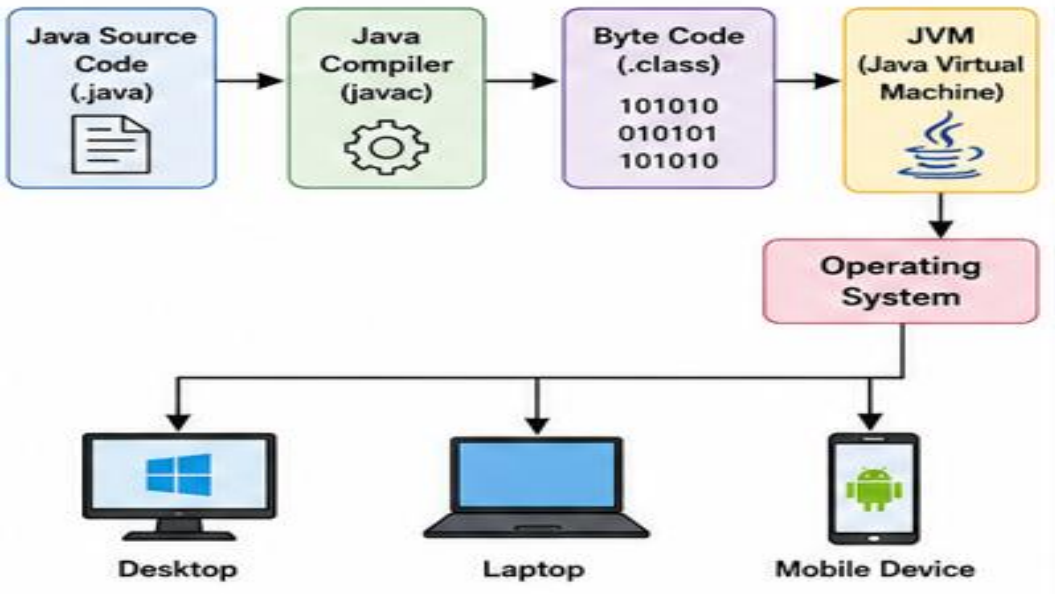
- Java is a high-level, object-oriented programming language used for developing mobile, web, and enterprise applications.
- It supports platform-independent application development using the Java Virtual Machine (JVM).

Features

- Platform independent.
- Object-oriented programming support.
- Secure and portable.
- Supports distributed applications.

How It Works

- Java source code is written by programmers.
- The Java compiler converts source code into bytecode.
- Bytecode runs on JVM present in different operating systems.
- JVM interprets and executes the program on the device.



Applications

- Android applications
- Web applications
- Enterprise systems

Real-Time Example

- Android mobile applications developed using Java.

BREW

Definition

- BREW (Binary Runtime Environment for Wireless) is a mobile application development platform developed by Qualcomm for wireless devices.

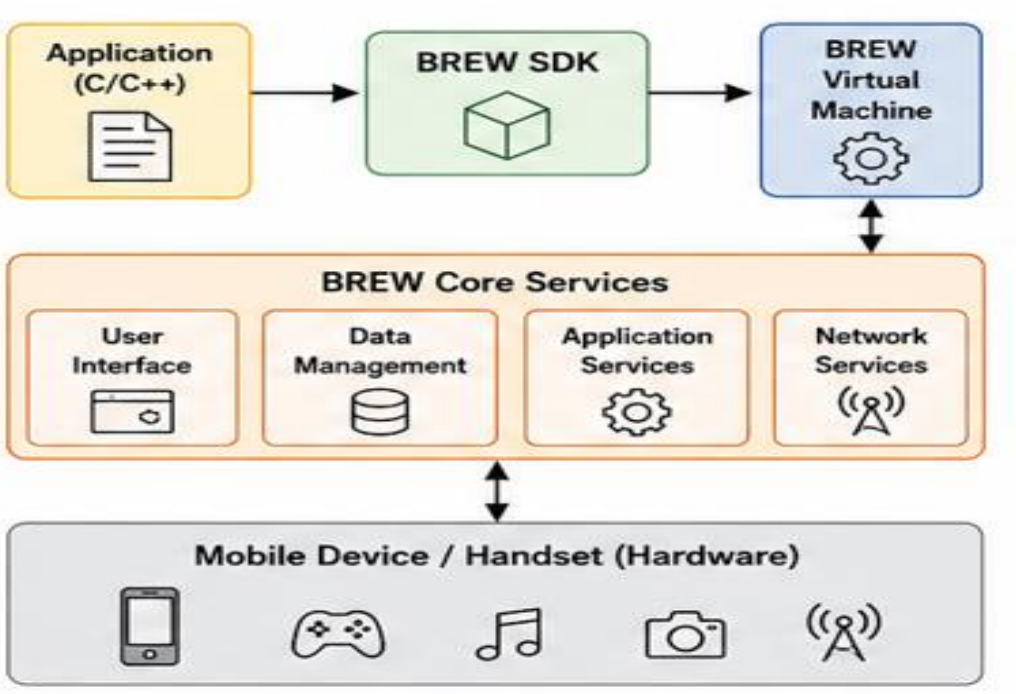
Features

- Supports wireless applications.

- Provides runtime environment for mobile apps.
- Supports multimedia applications.
- Efficient for embedded systems.

How It Works

- Applications are developed using C or C++.
- BREW runtime environment executes applications on mobile devices.
- It interacts with mobile hardware and wireless network services.
- Users can download and run applications through BREW-enabled phones.



Applications

- Mobile games
- Multimedia applications
- Wireless communication apps

Real-Time Example

- Early Qualcomm mobile phones using BREW applications.

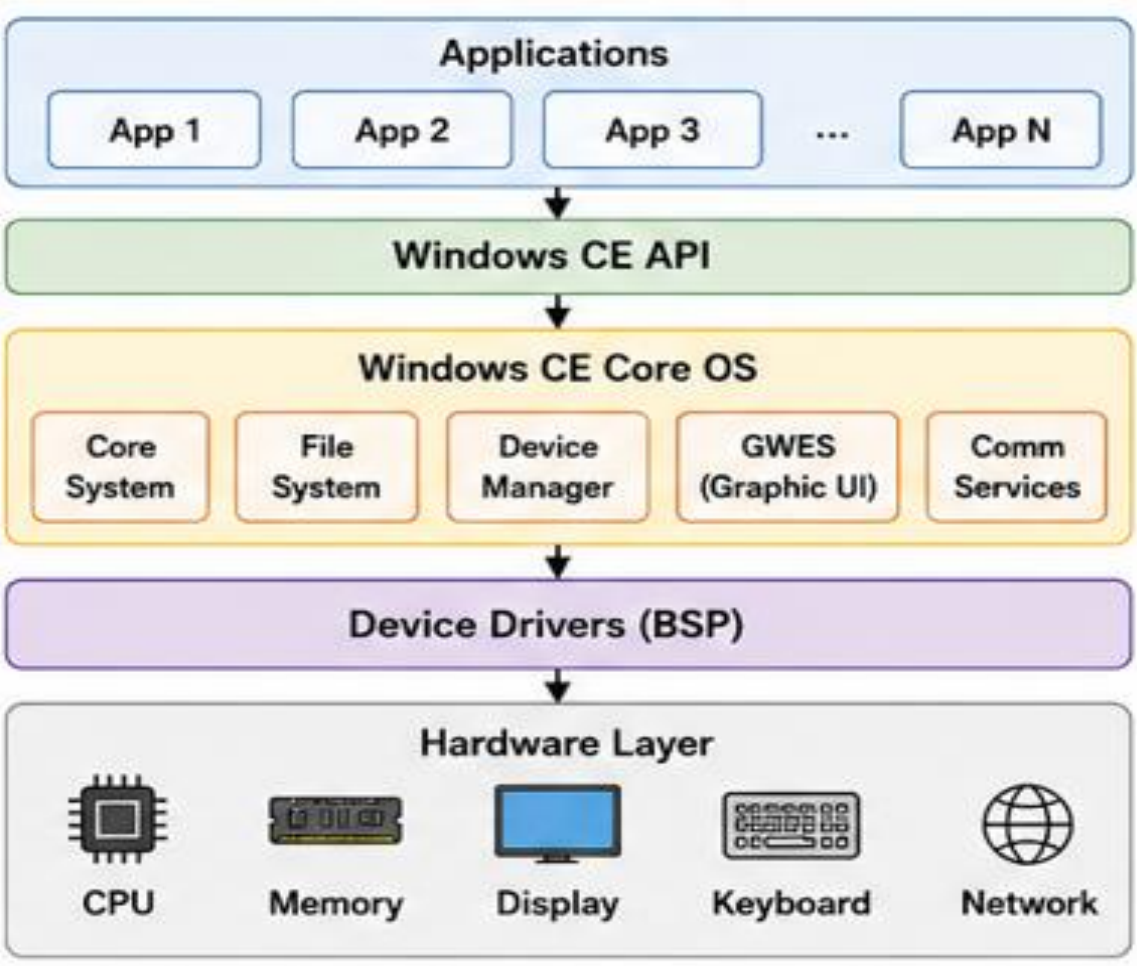
Windows CE

Definition

- Windows CE is a lightweight embedded operating system developed by Microsoft for handheld and mobile devices.

How It Works

- Windows CE operates on embedded and portable devices.
- It manages hardware resources, applications, and memory.
- Applications communicate with the OS through APIs.
- The operating system handles multitasking and device control.



Applications

- Embedded systems

- Handheld devices
- Industrial automation systems

Real-Time Example

- Portable medical devices using Windows CE.

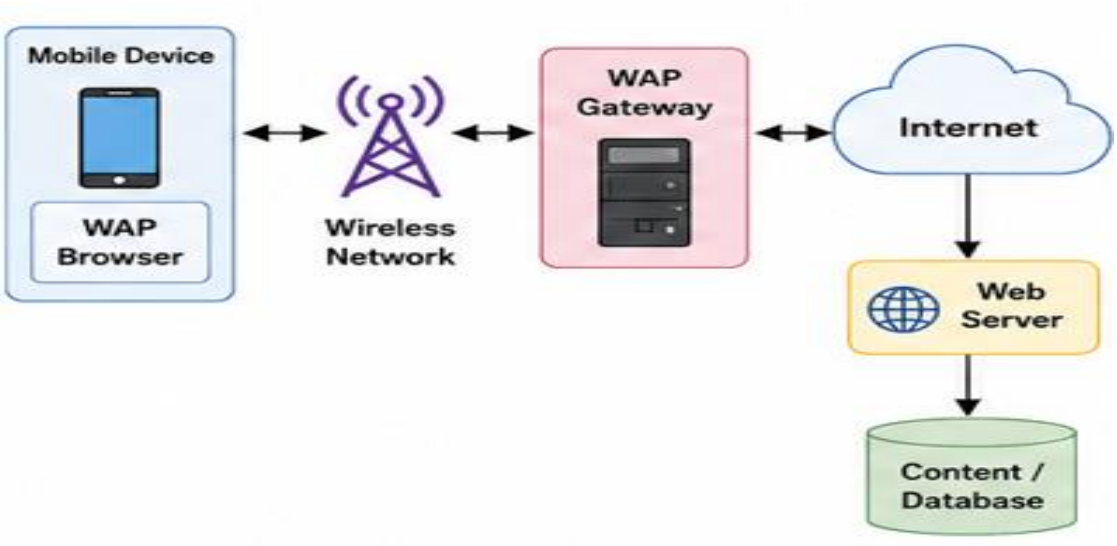
WAP

Definition

- WAP (Wireless Application Protocol) is a communication protocol used for accessing internet services on mobile devices.

How It Works

- Mobile device sends request through wireless network.
- WAP gateway converts mobile requests into internet-compatible format.
- Web server processes request and sends response back.
- Mobile browser displays web content to users.



Applications

- Mobile internet browsing
- Mobile banking
- Online information services

Real-Time Example

- Early mobile phones using WAP browsers for internet access.

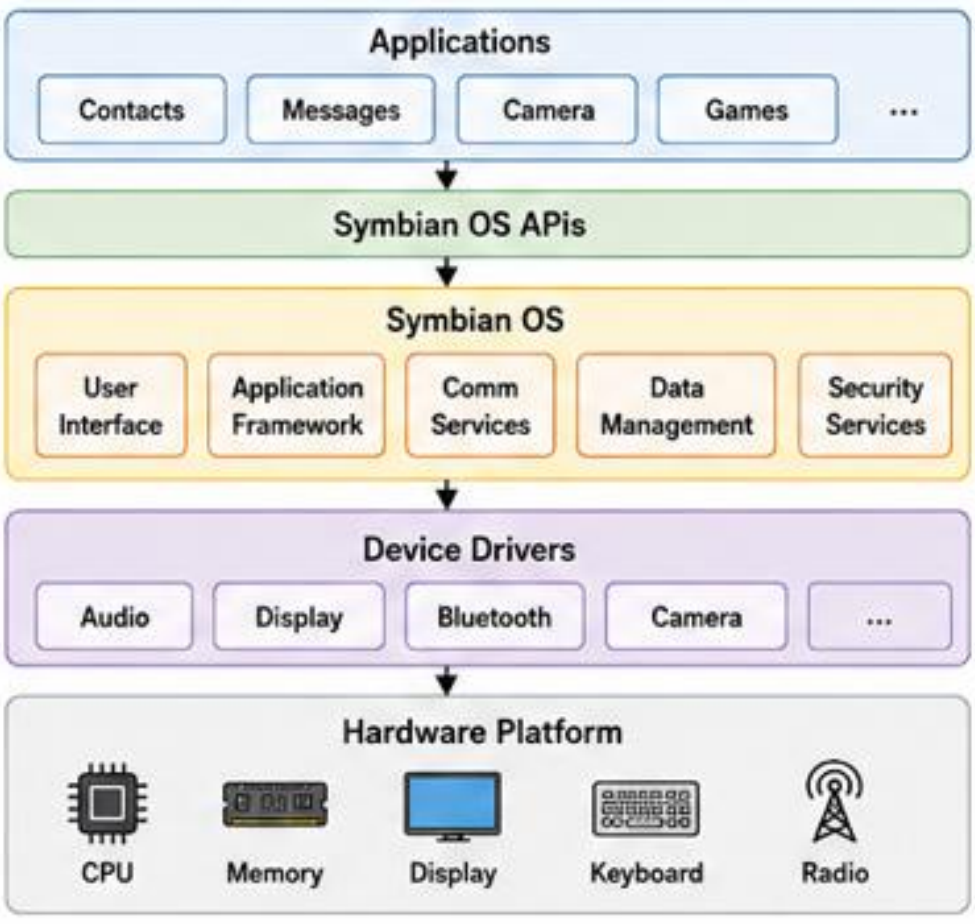
Symbian

Definition

- Symbian is a mobile operating system designed for smartphones and handheld devices.

How It Works

- Symbian manages mobile hardware and software resources.
- Applications run on the Symbian operating system.
- It handles communication, multitasking, and memory management.
- Users interact with applications through graphical interface.



Applications

- Smartphones
- Mobile multimedia applications
- Communication systems

Real-Time Example

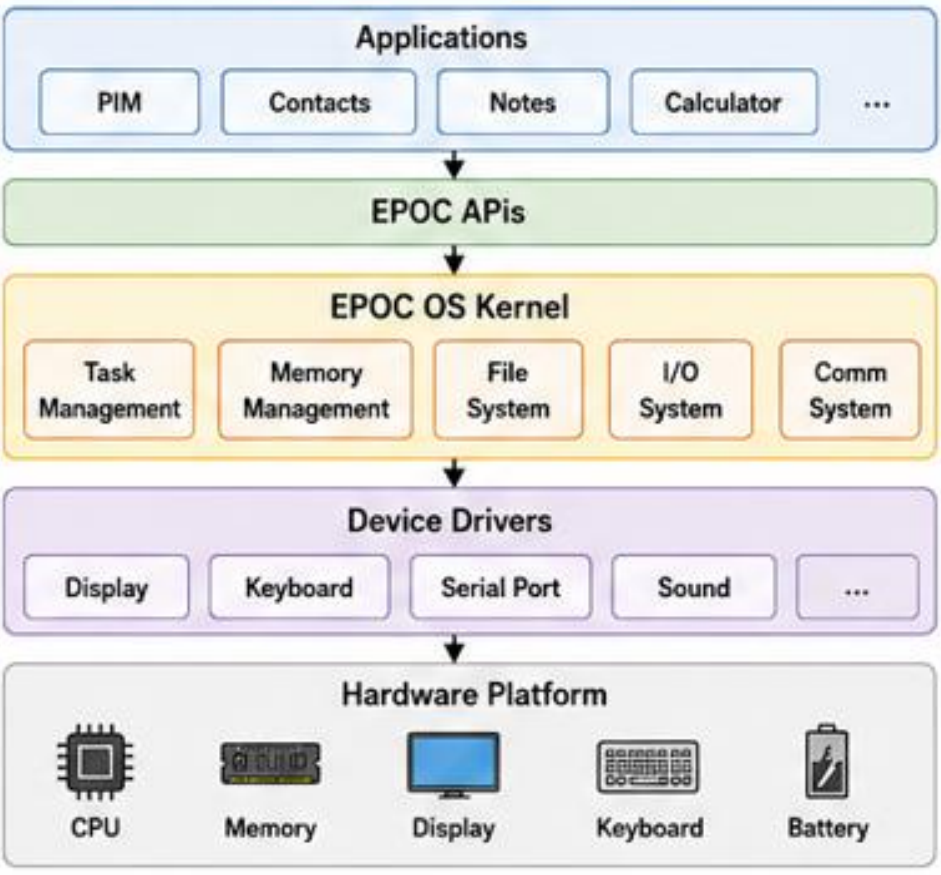
- Nokia smartphones using Symbian OS.

EPOC

Definition

- EPOC is a mobile operating system developed by Psion for handheld and portable devices.
- It later evolved into the Symbian operating system.

How It Works



- EPOC manages hardware and application execution in portable devices.
- It provides user interface and multitasking support.
- Applications communicate with system resources through the operating system.
- It controls memory, processing, and device communication.

Applications

- Personal digital assistants (PDAs)
- Handheld devices
- Mobile computing systems

Real-Time Example

- Psion handheld devices using EPOC operating system.